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A tissue boundary orchestrates the segregation of inner ear sensory organs

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eLife Assessment

This is an **important** study describing the morphological changes during boundary formation between sensory and non-sensory tissues of the inner ear. The authors provided **solid** evidence that a transcription factor, Lmx1a and ROCK-dependent actinomyosin are key for border formation in the inner ear. However, future studies will be needed to investigate the direct relationships among boundary formation, Lmx1a and ROCK. This work will be of interest to developmental biologists interested in boundary formation.

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Abstract

The inner ear contains distinct sensory organs, produced sequentially by segregation from a large sensory-competent domain in the developing otic vesicle. To understand the mechanistic basis of this process, we investigated the changes in prosensory cell patterning, proliferation and character during the segregation of some of the vestibular organs in the mouse and chicken otic vesicle. We discovered a specialized boundary domain, located at the interface of segregating organs. It is composed of prosensory cells that gradually enlarge, elongate and are ultimately diverted from a prosensory fate. Strikingly, the boundary cells align their apical borders and constrict basally at the interface of cells expressing or not the Lmx1a transcription factor, an orthologue of drosophila Apterous. The boundary domain is absent in *Lmx1a*-deficient mice, which exhibit defects in sensory organ segregation, and is disrupted by the inhibition of ROCK-dependent actomyosin contractility. Altogether, our results suggest that actomyosin-dependent tissue boundaries ensure the proper separation of inner ear sensory organs and uncover striking homologies between this process and the compartmentalization of the drosophila wing disc by lineage-restricted boundaries.

Introduction

The inner ear of vertebrates is composed of a series of liquid-filled chambers containing distinct sensory organs. Each organ contains a sensory epithelium populated with mechanosensory 'hair cells', separated from one another by supporting cells. In amniotes, the ventral part of the inner ear forms the cochlea, which hosts an auditory epithelium called the basilar papilla in birds and crocodiles, or the organ of Corti in mammals. The dorsal part forms the vestibular system, with three cristae (anterior, posterior and lateral) and their associated semi-circular canals responding to the angular rotations of the head, and the maculae of the utricle and saccule, sensitive to gravity and linear acceleration. These sensory organs are separated from one another by non-sensory territories with essential roles in inner ear function and homeostasis (Ekdale, 2016).

The development of the inner ear is a remarkable example of 3D tissue morphogenesis that starts with the formation of the otic placode, an epithelial thickening located on both sides of the hindbrain. The placode rapidly invaginates into the underlying mesenchyme to form the otic cup. Subsequently, the cup pinches off from the surface ectoderm and closes into the otic vesicle (or otocyst), which then undergoes a drastic remodelling to give rise to the inner ear (reviewed in Alsina and Whitfield, 2017; Basch et al., 2016a).

How are the different cell types and territories of the inner ear specified and positioned during its embryonic development? The prevalent view is that diffusible signals (eg Wnt, Hedgehog, FGF, Retinoic Acid) secreted by tissues surrounding the otocyst direct the expression of key factors (transcription factors and signalling molecules) to specify various otic cell identities (reviewed in Fekete and Wu, 2002; Ohta and Schoenwolf, 2018). It was also proposed that once specified, adjacent but distinct populations of otic progenitors might be prevented from intermixing by lineage-restricted tissue boundaries akin to those discovered at the interface of "embryonic compartments" in the Drosophila wing disc (Brigande et al., 2000b; Fekete, 1996). This 'compartment boundary' model has received some support from the analysis of mouse mutants exhibiting defects in restricted domains of the inner ear and some fate map studies, but it remains hypothetical with respect to the existence and precise location of tissue boundaries (Kil & Collazo, 2002).

One aspect of inner ear development that results in the spatial separation of adjacent groups of cells is the formation of its sensory organs. In the chicken or mouse otic vesicle, the prospective sensory domains, or prosensory domains, are recognizable as thickened epithelial patches in the ventro-medial and posterior regions (Knowlton, 1967). The prosensory patches express the Notch ligand Jag1, which promotes the maintenance of the prosensory fate by Notch-mediated lateral induction (reviewed in Daudet and Żak, 2020; Kiernan, 2013). They also express the High-Mobility Group transcription factor Sox2, which is essential for the formation of all inner ear sensory organs (Dabdoub et al., 2008; Kiernan et al., 2005) as well as otic neurogenesis (Steevens et al., 2019, 2017). At early developmental stages, Sox2 marks a wide antero-posterior "pan-sensory" domain occupying the ventro-medial wall of the otocyst (Steevens et al., 2019), due at least in part to the inhibition of Sox2 expression by Wnt activity in the dorsal aspect of the otocyst (Żak and Daudet, 2021). Subsequently, the pan-sensory domain splits into the posterior crista, and several sensory organs in the anterior part of the otocyst: the anterior crista, the lateral crista, and the utricle (Mann et al., 2017; Sánchez-Guardado et al., 2013). This segregation process was first described in the inner ear of fish and amphibians more than a century ago (Norris, 1892) and could have played an essential role in the evolution of the vertebrate inner ear, by allowing new sensory organs to acquire specific functions (reviewed in Fritzsch et al., 2002). Genetic studies in the mouse have shown that mutations or absence of specific transcription factors and signalling molecules can lead to defects in sensory organ segregation (reviewed in Alsina and Whitfield, 2017). One such factor is Lmx1a (cLmx1b in the chicken), a LIM-homeodomain transcription factor whose orthologue Apterous acts as a “selector gene” for the dorsal compartment of the Drosophila wing disc (Rincon-Limas et al., 1999). In the mouse inner ear, Lmx1a is expressed in the non-sensory territories separating the sensory organs, and its absence causes severe morphogenesis defects and partial fusions between adjacent sensory territories (Koo et al., 2009; Nichols et al., 2008; Steffes et al., 2012). Our previous work has shown that Lmx1a antagonizes Jag1/Notch signalling within some early sensory-competent cells, thereby leading those cells to adopt a non-sensory fate between adjacent sensory organs (reviewed in Daudet and Żak, 2020; Mann et al., 2017). Live-imaging studies in zebrafish embryos have also shown that a progressive reduction in Jag1b expression occurs within the cells located between segregating sensory organs, confirming the central role played by the regulation of Notch activity in this process (Ma & Zhang, 2015). Despite these insights into the genetic regulation of sensory organ segregation, the cellular basis of this process is still unclear. Is it solely dependent on changes in cell character regulated by Notch and Lmx1a, or does it involve other processes and if so, of which type?

To tackle this question, we examined the changes in cell morphology during the segregation of the lateral (LC) and anterior (AC) cristae from the pan-sensory domain in the embryonic chicken and mouse otocyst. We identified a population of prosensory cells with enlarged surfaces, which appear at the interface of segregating organs and gradually elongate and re-align along the border of the prospective cristae before down-regulating Sox2 expression. As segregation proceeds, these large cells constrict basally at the interface of Lmx1a-positive (on the cristae side) and negative (on the pan-sensory domain side) cells, strongly suggesting the formation of a lineage-restricted tissue boundary. In Lmx1a-null mouse otocysts, the boundary domain is disrupted; the AC and the LC initiate their segregation from the utricle but they remain fused to one another, indicating that Lmx1a is required for the differentiation of the non-sensory cells separating the two cristae. Finally, we show that the perturbation of ROCK-dependent actomyosin contractility, a common player in the establishment of lineage-restricted boundaries, leads to defects in the organisation of the boundary domain and abnormal sensory organ segregation.

Altogether, our results strongly suggest that a lineage-restricted tissue boundary forms at the interface of segregating sensory organs, providing support to the compartment-boundary model of inner ear development. They also uncover striking similarities in the genetic circuits and cellular mechanisms of tissue segregation in the inner ear and the Drosophila wing disc.

Results

A specialized boundary domain forms between segregating vestibular organs in the chicken otocyst

We focused in this study on the formation of the anterior (AC) and lateral (LC) cristae, using whole-mount preparations of the otic vesicle immunostained for the prosensory marker Sox2. In the chicken inner ear, the AC is the first to segregate from the large anterior pan-sensory domain between stage HH23-25 (E4-E5), followed by the LC between HH25-27 (E4.5-E5.5) (n=10) (**Figure 1a-c**; see also Mann et al., 2017). When we examined at high magnification phalloidin-stained samples, we noticed that the cells located at the interface of the cristae and the pan-sensory domain had a larger apical surface than those of surrounding territories.

Focusing on the LC, which could be identified from the earliest stages of its formation, we distinguished 3 stages of segregation as a function of Sox2 expression and the extent of separation from the pan-sensory domain. At stage 1 (HH25-26), the prospective LC formed a budding that was not yet separated from the pan-sensory domain (n=4) (**Figure 1a** and **d-e’’**). Seven to nine rows of Sox2-positive cells located at the interface of the future LC and the pan-sensory domain appeared slightly larger than those located on either side. At stage 2 (HH26-27; **Figure 1b**, and **g-**

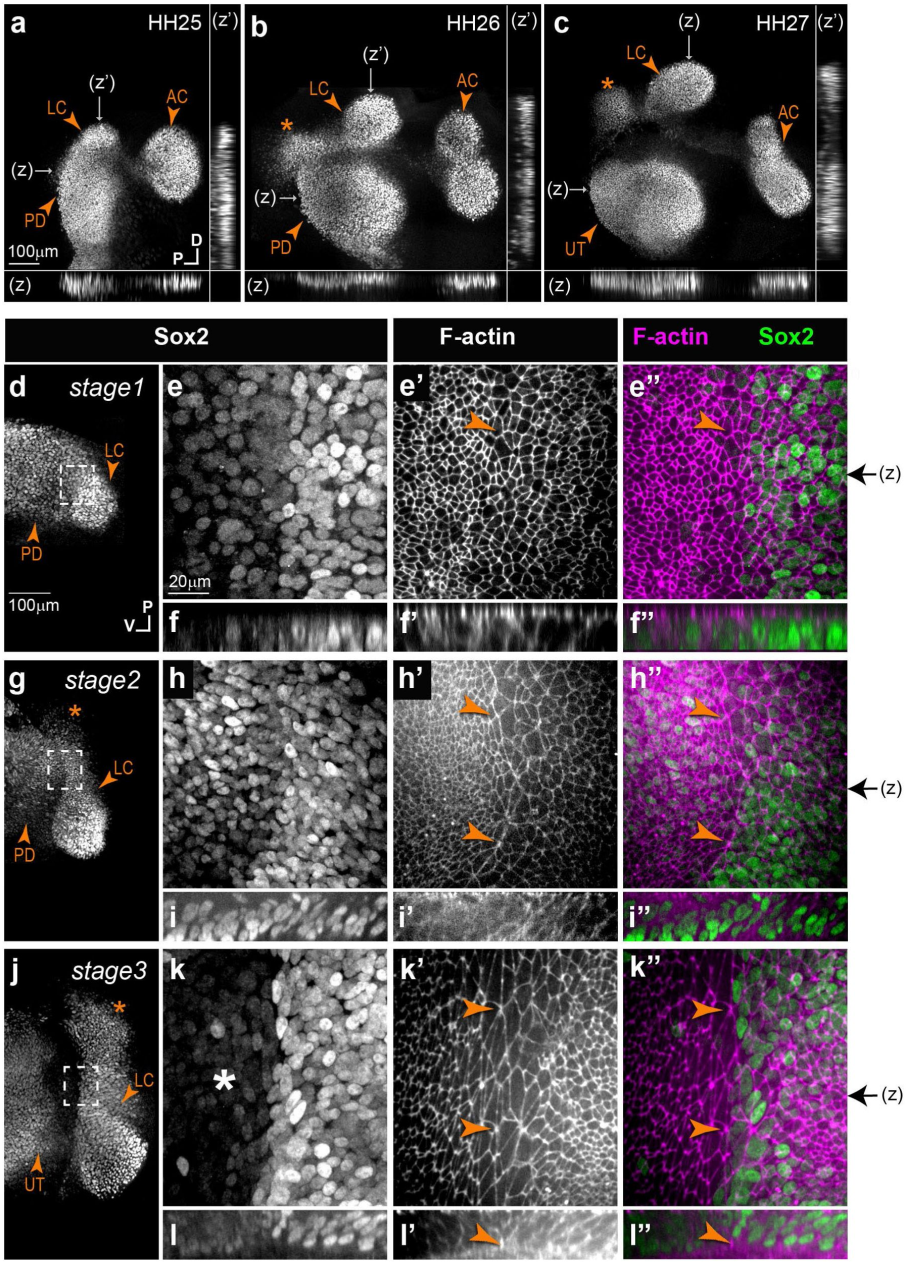


Figure 1.

Segregation of the anterior and lateral cristae in the chicken inner ear.

a-c) Low magnification surface views and transverse line optical reconstructions (z,z') of whole-mount preparations of the anterior part of the chicken otocyst at different stages of development (HH=Hamburger and Hamilton stages), immunostained for Sox2 expression. The anterior (AC) and lateral (LC) cristae segregate sequentially from the pan-sensory domain (PD), which then becomes the utricle (UT). The LC is connected to a second Sox2-expressing domain (stars) that does not give rise to a sensory organ. **d-l'')** Stages 1 (**d-f''**), 2 (**g-i''**) and 3 (**j-l''**) of the segregation of the LC, characterized by a progressive down-regulation of Sox2 expression in the domain separating the LC from the PD domain. High magnification views (**e-e''**, **h-h''** and **k-k''** correspond to the boxed areas in respectively **d-g-j)** reveal striking changes in cellular morphology and organization during LC segregation. A specialized boundary domain composed of large cells that progressively align their membranes along the border of the LC (arrowheads in **e'-l''**) becomes clearly visible from stage 2 onwards. This is concomitant to the formation of a basal constriction at the interface of the crista and pan-sensory domain, clearly visible at stage 3 in orthogonal z-reconstructions of the boundary domain (arrowheads in l-l”). Black arrows on the side of images e”, h”, and k” indicate the position of z-reconstructions.

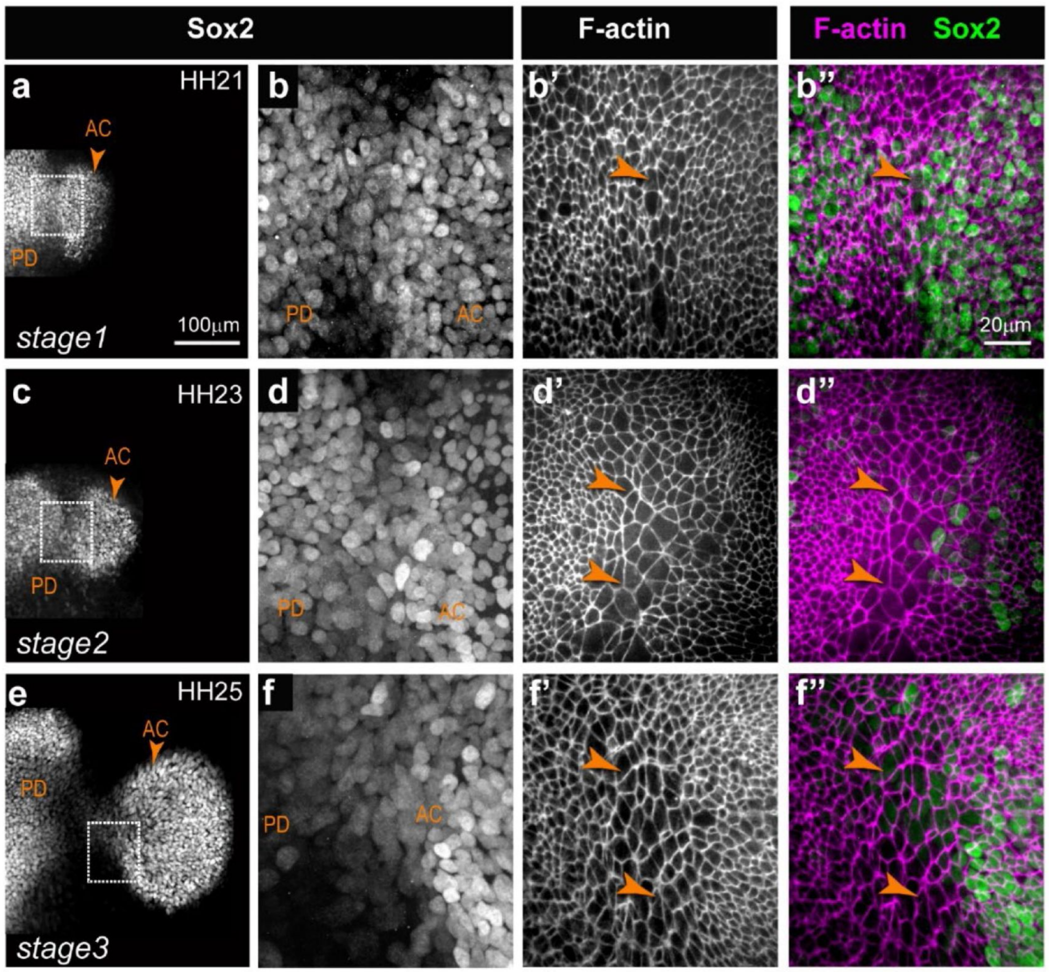


Figure 1 – Supplementary Figure 1.

Segregation of the anterior crista in the chicken inner ear.

Low and high magnification views of the anterior crista (AC) and pan-sensory domain (PD) at different stages of development (HH=Hamburger and Hamilton stages), immunostained for Sox2 expression and labelled with fluorescent phalloidin. At stage 1 of segregation (**a-b''**), the two patches are in contact and few cells have enlarged surfaces at their interface (arrowheads in **b'-b''**). At stage 2 (**c-d''**), several rows of larger cells are seen at the interface of the PD and prospective AC (arrowheads in **d'-d''**). At stage 3 (**e-f''**), the two patches are separated by a Sox2-negative domain and several rows of large cells (arrowheads in **f'-f''**).

h''), Sox2 started to be downregulated within the interface domain separating the LC from the pan-sensory domain (n=4). Scattered Sox2-positive nuclei were also present at the edge of a second Sox2-positive lobe of the LC, which remained connected to the pan-sensory domain. Of note, the second lobe of the LC does not give rise to sensory cells at later developmental stages and as discussed later, it is not present in the mouse inner ear. In some cases, the large cells appeared to align their apical borders along the edge of the crista, forming a multi-cellular actin cable-like structure suggesting a coordinated increase in cell bond tension (**Figure 1g-h**”). At stage 3 (HH27-28; **Figure 1c** and **j-k''**), the two lobes of the LC were clearly separated from the pan-sensory domain by an 80-180 µm wide territory containing cells with no (or very reduced) Sox2 expression (n=4). At the border of the LC, 7-10 rows of large cells remained visible. Comparable changes were observed at the border of the AC (**Figure 1-Supplementary Figure 1**), although we were not able to analyze the early stages of segregation for this organ.

Concomitant to these changes in cell surface morphology, we noticed a progressive enrichment in F-actin staining at the base of the epithelium along the presumptive border of the LC (n=4) (**Figure 1f-f”, i-i”, l-l”** and **Figure 2**). At stages 2-3, it was associated to a clear basal constriction of the cells and an upward displacement of the basal lamina along the presumptive border of the LC (**Figure 1-l”** and **Figure 2 b-c**). This suggests that the basal constriction could at least partly explain the enlarged apical surface of the cells composing the specialized boundary domain.

We next used the Epitool (Heller et al., 2016) and Icy softwares (see Methods) to analyse in greater detail the changes in surface cell morphology during the segregation of the LC. Cell skeletons were generated from phalloidin-stained preparations (n=3 samples per stage), allowing the quantification of cell surfaces, elongation (the ratio between their long and short axis) and orientation (**Figure 3**). This analysis showed that the large cells at the interface of the LC and pan-sensory domain expanded their apical surface and increased in number from stage 1 to stage 3 (**Figure 3b-c**); they became more elongated than neighbouring cells from stage 2 (**Figure 3d**) and their long axis became gradually aligned along the same planar polarity vector, parallel to the edge of the crista, at stage 2-3 (**Figure 3e-f**). The increase in cell surface area was maximum in the interface domain but observed throughout the tissue, explaining the overall reduction in cell density observed between stage 1 and stage 3. The elongation and re-orientation of the cells was also noticeable throughout the epithelium, suggesting that these are primarily driven by global anisotropic changes in tissue tension. As seen in LC samples stained for the apical surface marker ZO-1 and DAPI (**Figure 3 -Supplementary Figure 1a-a”**), the majority of cells with a large surface area were not undergoing mitosis and their nuclei had a comparable size to those of neighbouring prosensory and non-sensory cells Altogether these results indicate that very dynamic changes in cell morphology and tissue patterning occur at the interface of the prospective LC and the pan-sensory domain before (stage 1) and during (stage 2-3) their segregation.

A specialized boundary domain forms at the interface of Lmx1a-positive and Lmx1a-negative cells in the mouse otocyst

To find out if a comparable interface domain forms in the mammalian inner ear, we analysed the otocysts of Lmx1a^{GFP} knock-in mouse embryos. In a previous study, we showed that the inner ear of heterozygous Lmx1a$^{GFP/+}$ mice develops normally and GFP expression becomes gradually upregulated in the non-sensory territories surrounding the vestibular prosensory patches (Mann et al., 2017). However, we did not investigate in detail the earliest stages of segregation of the two cristae from the pan-sensory domain.

In E11.5 otocysts (n=4), two small Sox2-positive patches corresponding to the prospective AC and LC were visible at the anterior edge of the pan-sensory domain (**Figure 4a**). However, in contrast to what was observed in the chicken otocyst, the LC did not have a second Sox2-positive

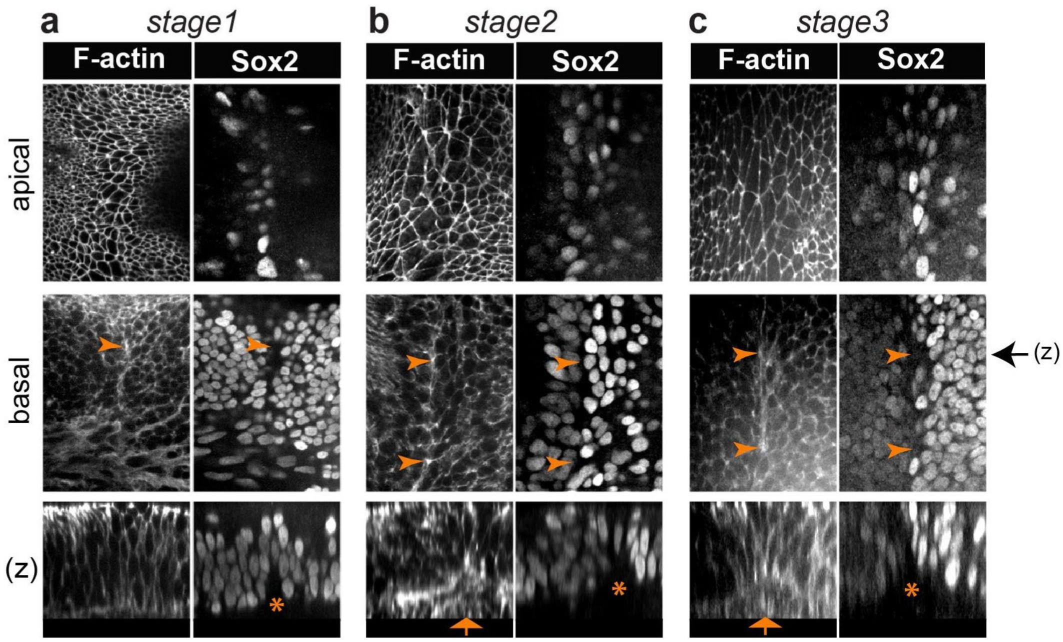


Figure 2.

A multicellular basal constriction forms at the interface of the lateral crista and pan-sensory domain.

a-c) Samples stained for F-actin and Sox2 expression at stages 1-3 of LC segregation in the chicken inner ear. Optical sections collected from the surface and basal planes, and orthogonal reconstructions (z). In all images, the pan-sensory domain is on the left and the LC on the right. **a)** F-actin is enriched at the base of the interface of the LC from stage 1 (arrowheads). **b-c)** At stage 2-3, the F-actin enrichment becomes very clear along the prospective border of the LC (arrowheads). A basal constriction of the cells (arrows in z) and an upward displacement of the basal lamina (stars) are also noticeable. Black arrow on the side of images of the basal plane indicates the corresponding positions of z-reconstructions.

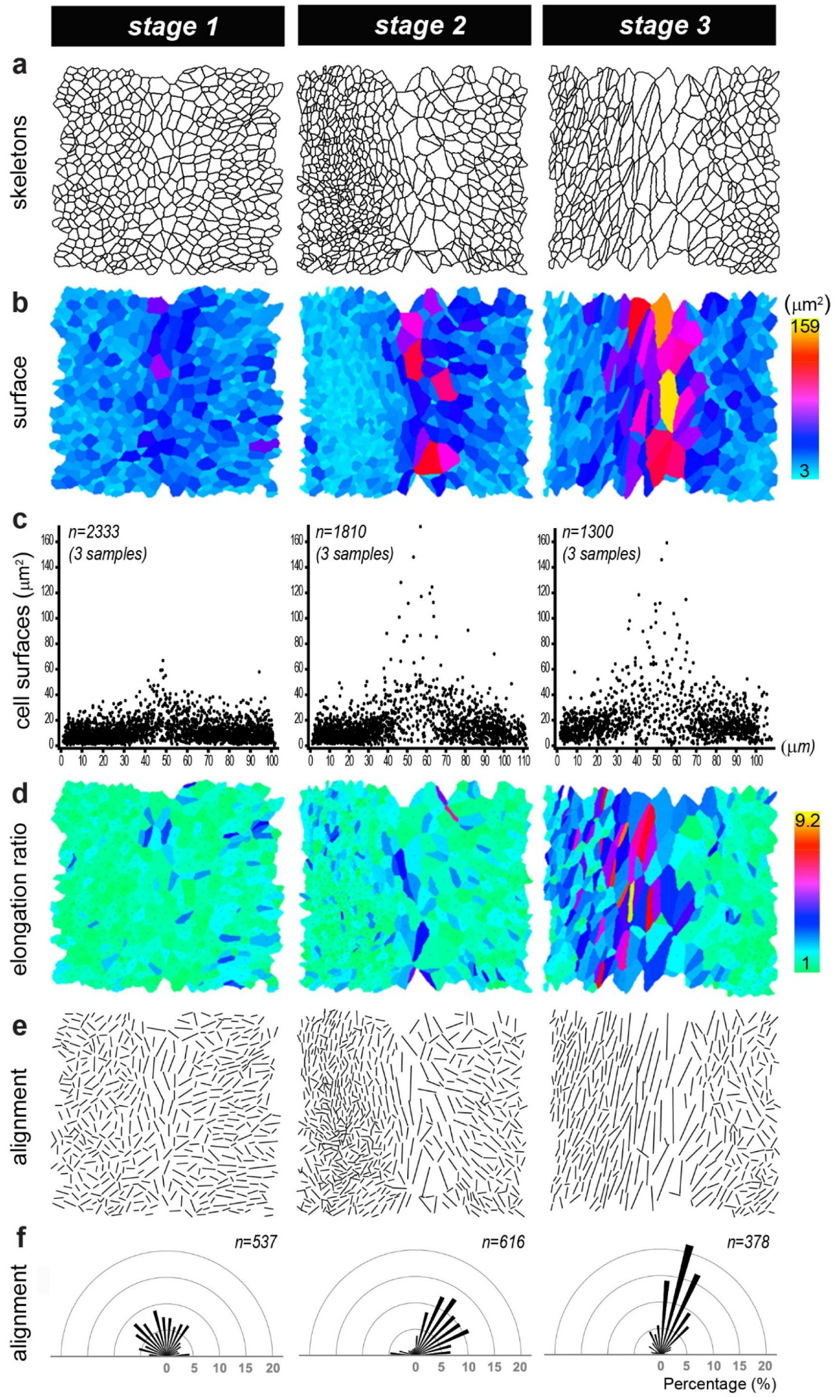


Figure 3.

Analysis of the morphological changes during segregation of the lateral crista in the chicken otocyst.

a) Cell skeletons generated from phalloidin stained preparations at three stages of LC segregation. In all images, the interface domain is in the middle, the pan-sensory domain on the left, and the LC on the right. **b-c**) Colour scale and pooled scatter plots of the surface cell area show that the largest cells are located at the interface of the LC and pan-sensory domain from stage 1. **d**) Color-coded representation of the cell elongation ratio. **e**) Visualization of the long axis of the cells showing a progressive elongation between stages 1-3 and a clear alignment of the cells along the border of the LC at stage 3. **f**) Angular distribution of cell orientations in the boundary domain shown in e relative to the longitudinal axis of the boundary domain. “n” represents the number of cells analysed for each sample.

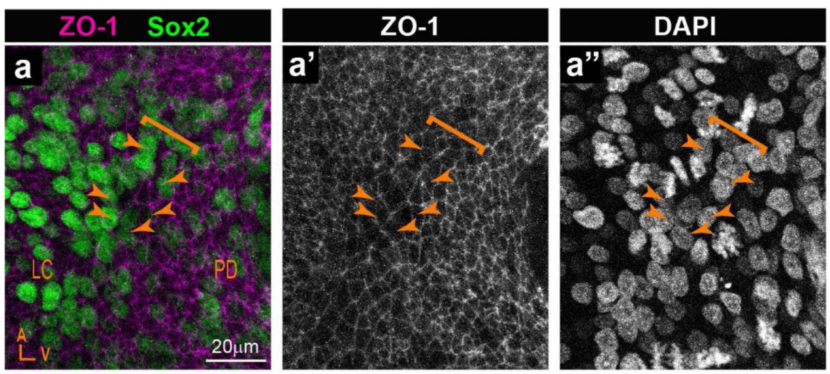


Figure 3 - Supplementary Figure 1.

Visualization of cell nuclei at the boundaries of segregating sensory organs.

A-a”) Whole-mount surface view of a LC boundary stained for the apical surface marker ZO-1, Sox2, and DAPI. DAPI staining in **a”** shows no apparent difference in the sizes of cell nuclei between the cells with large surface (**arrowheads**) at the boundary (**brackets**) and the neighbouring cells.

lobe and the segregation of the AC and LC was synchronous. In fact, the AC was still connected to the pan-sensory domain at the time the LC budding emerged from it. In some way, the size and location of the AC and LC in E11.5 mouse otocysts resembled that of the two Sox2-positive lobes of the LC in the chicken otic vesicle at stage 2 (compare with **Figure 1b**). At this stage, Lmx1a/GFP expression pattern overlapped with that of Sox2 in between the presumptive cristae and at their interface with the large pan-sensory domain (**Figure 4b-b**'). Seen from the surface, the border of the Lmx1a-GFP positive territory was relatively jagged, although GFP-positive and negative cells remain clearly separated. The interface domain contained cells with a slightly enlarged surface in 1 out of 4 samples only. In 2 out of 4 E11.5 samples with good quality phalloidin staining, we observed segments of basal F-actin enrichment at the edge of the GFP-positive domain (**Figure 4c-c'** and **Figure 4 -Supplementary Figure 1a**). We conclude that E11.5 corresponds to an early stage of segregation of the cristae, equivalent to 'stage1' in the chicken inner ear.

At E13.5 (n=5), Sox2 expression marked 3 discrete patches corresponding to the AC, the LC and the prospective utricle (**Figure 4d**). Lmx1a/GFP expression was still present in Sox2-positive cells at the border of the AC and LC and in prospective non-sensory cells in between the two cristae which exhibited a clear reduction in Sox2 expression (**Figure 4e-e**'). Furthermore, an interface domain with enlarged cells was clearly recognizable in between the cristae and prospective utricle (**Figure 4f-f'** and **Figure 4 - Supplementary Figure 1b**). Remarkably, Lmx1a-GFP was detected in the half of the large cell domain facing the cristae; the other half, facing the prospective utricle, had no or very low levels of GFP fluorescence and exhibited reduced Sox2 expression. The apical membranes of the cells at the interface of GFP-positive and negative domains were in some samples markedly aligned (**Figure 4 - Supplementary Figure 2**), suggesting an increase in cell bond tension along the border of Lmx1a expression.

In the basal planes, F-actin staining was enriched in multicellular cable-like structures and a basal constriction similar to that observed in the chicken otocyst (at stage 2-3) was occasionally present at the precise interface of the Lmx1a-expressing and non-expressing cells (orthogonal reconstructions in **Figure 4f-f'** and **Figure 4 - Supplementary Figure 1b**).

In neonate mice (n=3; **Figure 4g**), Lmx1a/GFP expression persisted in a wide non-sensory territory separating the two cristae from the utricle. In contrast to earlier stages, there was no overlap between Sox2 and GFP expression at the border of the sensory organs (**Figure 4h-h**') and a basal constriction was either absent or impossible to precisely locate in the samples examined (**Figure 4i- i'**).

Altogether, these data indicate that the cellular correlates of cristae segregation are comparable in the mouse and chicken inner ear, despite some differences in its dynamics and outcomes. In both species, a specialized population of boundary cells with enlarged apical surfaces appears at the interface of the two cristae and the utricle at early stages of their segregation. In the apical plane of the epithelium, cells elongate and align their borders in the middle of the large cell domain, at the precise interface of Lmx1a-expressing and non-expressing cells. In the basal planes, the interface of Lmx1a-GFP and positive cells is marked by the formation of multicellular actin-cable-like structures, and in some cases a basal constriction and upward displacement of the basal lamina. This strongly suggests that a lineage-restricted tissue boundary separates these two cell populations.

The boundary domain is disrupted in *Lmx1a*-null mice

The complete absence of Lmx1a function leads to severe defects in inner ear morphogenesis, including absence of semi-circular canals or tissue constrictions and an abnormal growth and segregation of sensory organs (Koo et al., 2009; Mann et al., 2017; Nichols et al., 2008; Steffes et al., 2012). To find out if Lmx1a is required for the formation of the boundary domain, we compared cell morphology and organisation in E14 *Lmx1a*$^{GFP/+}$ heterozygous (n=8) versus *Lmx1a*$^{GFP/GFP}$ homozygous (*Lmx1a*-null; n=5) knock-in mice (**Figure 5**). Otocysts from

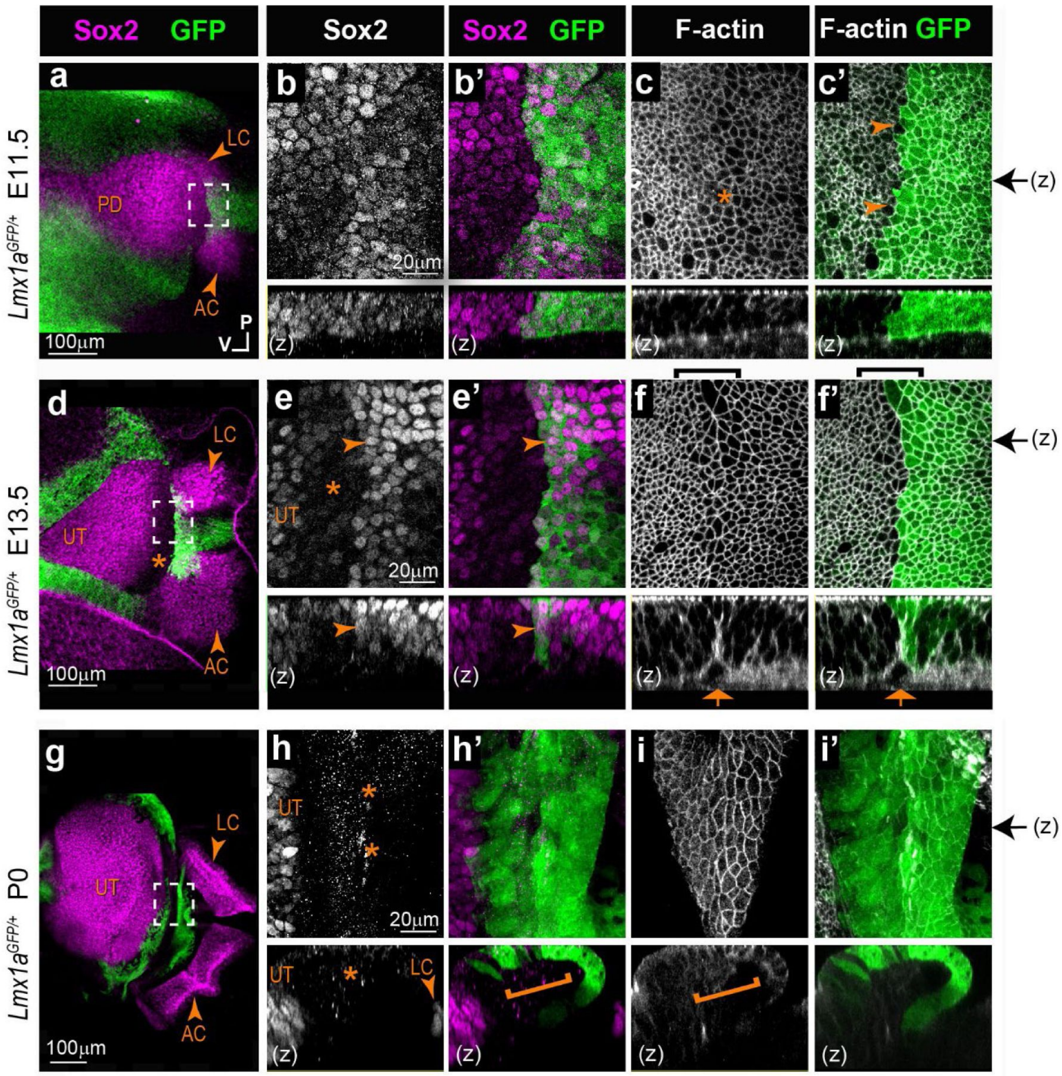


Figure 4.

Segregation of the anterior and lateral cristae from the utricle in the heterozygous *Lmx1a* *GFP/+* knock-in mouse inner ear. Whole-mount surface views and orthogonal reconstructions

(z) of the anterior part of the otocysts from *Lmx1a* $^{GFP/+}$ mouse at different stage of development, immunostained for Sox2 expression and stained for F-actin. **a)** At E11.5, the prospective anterior (AC) and lateral (LC) cristae are visible at the anterior edge of the pan-sensory domain (PD). Lmx1a/GFP expression pattern overlaps with that of Sox2 in between the presumptive cristae and at their interface with the PD **(b-b')**. The border of the Lmx1a-GFP positive territory is jagged, but GFP-positive and negative cells are clearly separated **(arrowheads in c')**. Cell surfaces appear slightly enlarged at the limit of the Lmx1a/GFP **(star in c). d)** At E13.5, the AC, LC and UT are separated. Lmx1a/GFP expression is present in Sox2-positive cells at the border of the AC and LC **(arrowheads in e-e')** and in prospective non-sensory cells in between the two cristae, which exhibited a clear reduction in Sox2 expression. Cells with enlarged surface areas are clearly recognizable in between the LC and UT **(brackets in f-f')**. Lmx1a/GFP is detected in the half of the large cell domain facing the cristae. A basal constriction is present at the precise interface of the Lmx1a-expressing and non-expressing cells **(arrows in f-f')**. At P0, Lmx1a/GFP expression persisted in a wide non-sensory territory separating the two cristae from the UT **(g)**. There is no overlap between Sox2 and GFP expression at the border of the sensory organs **(stars & bracket in h-h')**. No basal constriction is observed **(bracket in i)**. Black arrows on the side of images c', f', and i' indicate the position of z-reconstructions.

Figure 4 - Supplementary Figure 1.

Formation of a multicellular actin cable-like structure during segregation of the anterior and lateral cristae from the utricle in *Lmx1a* $^{GFP/+}$knock-in mouse.

a-b) Whole-mount views of surface and basal planes, as well as orthogonal reconstructions (z) of samples stained for F-actin and Sox2 expression. In all images, the developing utricle is on the left and the prospective anterior and lateral cristae are on the right. **a)** At E11.5, a basal F-actin enrichment is visible along a short segment of the interface between Lmx1a-GFP positive and negative cells (arrowheads). **b)** At E13.5, F-actin enrichment and an actin cable-like structure is seen in basal planes (arrowheads) and in z-reconstructions along the border of Lmx1a-GFP expression. However in this particular example, the upward displacement of the basal lamina is not visible or very moderate (star).

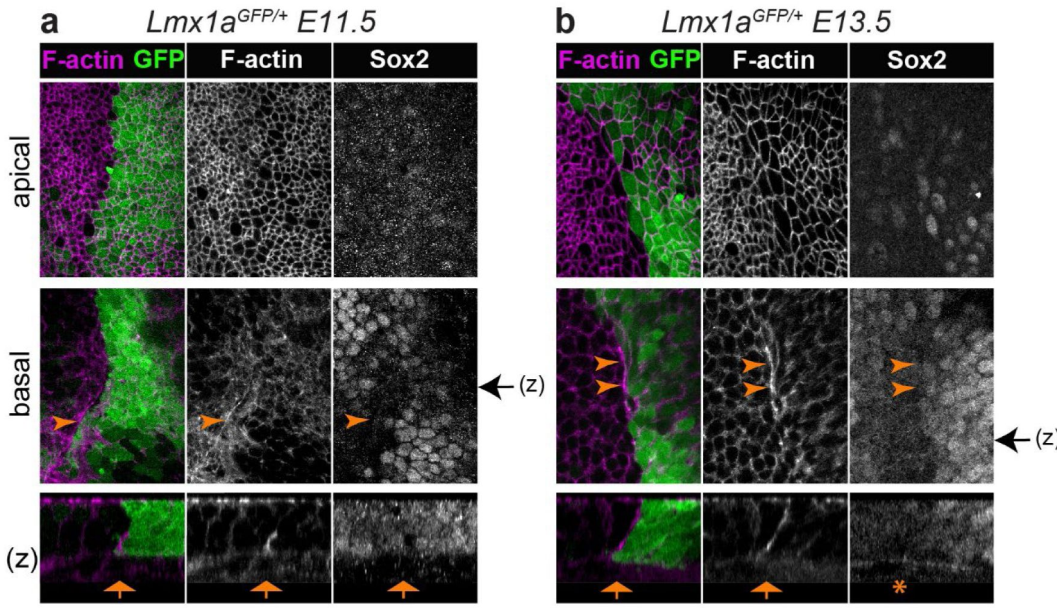
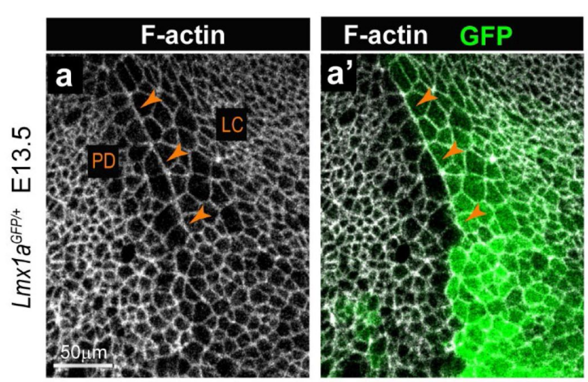


Figure 4 - Supplementary Figure 2.

Alignment of cell borders at the interface of Lmx1a-positive and negative cells in the mouse inner ear. (a-a')

Whole-mount preparations of the interface between the pan-sensory domain (PD) and lateral crista (LC) in an E13.5 *Lmx1a* $^{GFP/+}$ mouse otocyst. F-actin labelling reveals a marked alignment of apical cell borders along the boundary of Lmx1a-GFP expression (arrowheads).



homozygous *Lmx1a*$^{GFP/GFP}$ animals had very severe defects in the morphology of the AC and LC, which in all samples exhibited various degrees of expansion and fusion with one another (compare **Figure 5a** and **5h**). Consistent with our previous observations (Mann et al., 2017), Sox2 expression was maintained in the GFP-positive cells within the fused cristae, suggesting that the fusion of the two cristae is to a large extent due to a specification defect: in the absence of Lmx1a, sensory-competent cells fail to differentiate into non-sensory cells at the interface of the AC and LC, producing instead a large and continuous sensory territory. However, a relatively straight border of GFP-expression was present at the edge of the fused cristae (**Figure 5b** and **5i**) and a domain with reduced Sox2 expression was present between the cristae and utricle of the *Lmx1a*$^{GFP/GFP}$ samples, although not as clearly defined than in the *Lmx1a*$^{GFP/+}$ samples (stars in **Figure 5c-d** and **5j-k**). This suggests that the initial segregation of the crista and utricle pools of sensory progenitors is to some extent independent from Lmx1a function. Nevertheless, there were some differences in GFP pattern, cell morphology and organisation at the interface of the fused cristae and utricle. Firstly, GFP pattern was very mosaic across *Lmx1a*$^{GFP/GFP}$ cells, including at the border of the GFP-positive domain. This may be due to the abnormal differentiation of cells of the Lmx1a-lineage as well as increased mixing between cells belonging or not to the Lmx1a-lineage at the edge of the cristae. Secondly, there was no basal constriction at the interface of the GFP-positive and negative cells in the *Lmx1a*$^{GFP/GFP}$samples (compare **Figure 5f-g** and **5m-n**). Finally, the cellular organisation of the boundary domain was markedly different in the *Lmx1a*$^{GFP/GFP}$ samples. In fact, a quantitative analysis of cell surface morphology showed that cells on both sides of the boundary domain in *Lmx1a*$^{GFP/GFP}$ samples (n=3) tended to be larger, more aligned with one another and more elongated than in *Lmx1a*$^{GFP/+}$ ones (n=2) (**Figure 5 – Figure Supplementary 1**). Interestingly, these changes affected both GFP-positive and negative domains, suggesting non-cell autonomous consequences to the absence of *Lmx1a*.

Altogether, these results show that Lmx1a is required for the differentiation of the non-sensory territories between the AC and LC and the maintenance of the boundary domain between the cristae and the utricle. This is consistent with the notion that Lmx1a acts as a “selector” for the non-sensory fate in the domain separating the two cristae. Lmx1a function is also required for the formation of the large cell domain and its associated basal constriction at the limit of the Lmx1a-expressing domain, but neither appear necessary for the initial separation of the utricle from the cristae.

Pharmacological blockade of ROCK-dependent actomyosin contractility disrupts sensory organ segregation

In other contexts, tissue boundary formation or maintenance depends on increased actomyosin contractility at the interface of lineage-restricted compartments (reviewed in Dahmann et al., 2011; Monier et al., 2011). The resulting increase in cell bond tension is thought to create a physical ‘fence’ that restricts cell intermixing. To find out if this might also be the case in the chicken inner ear, we investigated the pattern of actomyosin activity using immunostaining for anti-phospho-myosin light chain 2 (PMLC) (**Figure 6**). The staining for PMLC was in general relatively weak and diffuse throughout epithelial cells, but was markedly increased and punctate within Sox2-expressing prosensory domains. The boundary between segregating sensory organs also exhibited strong PMLC punctate staining extending from the surface to the base of the epithelium (n=7). We then set up free-floating organotypic cultures of HH25 chicken otocysts, an early stage of segregation of the LC, and treated these with Y-27632, a potent inhibitor of ROCK-dependent actomyosin contractility (**Figure 7**).

After short-term (3h) culture in control DMEM medium (n=3; **Figure 7a-c**), HH26 otocysts at early stages of LC segregation had a well-recognizable large cell domain, but only one out of three samples examined had a clear basal constriction. Multicellular rosettes were frequently observed at the surface of the epithelium (**Figure 7b**’). However, the interface between the Sox2-positive

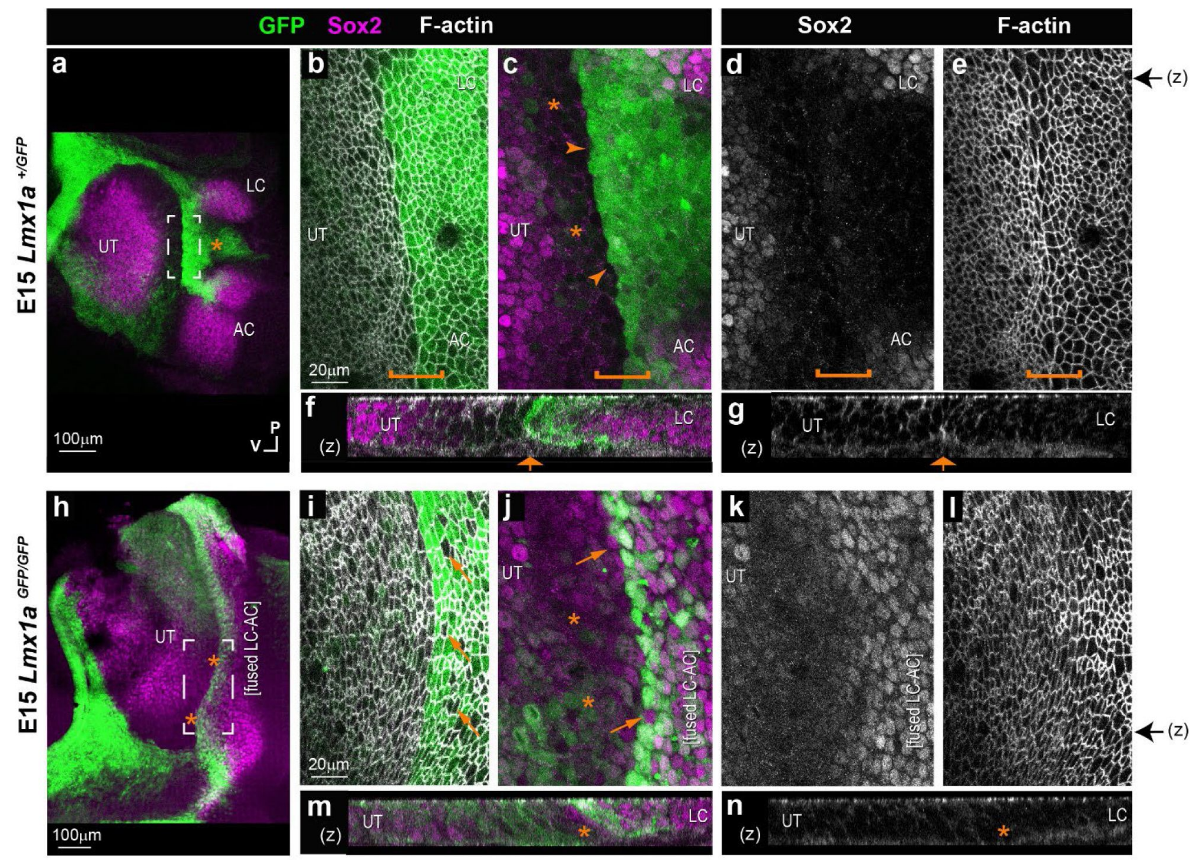


Figure 5

Comparison of the boundary domain in *Lmx1a* $^{GFP/+}$ and *Lmx1a* $^{GFP/GFP}$ mouse inner ear.

Whole-mount preparations of the utricle (UT), lateral (LC) and anterior (AC) cristae region, after immunostaining for Sox2 (magenta) and F-actin labelling with phalloidin (white). **(a)** In the E15 *Lmx1a* $^{GFP/+}$ inner ear (**a-g**), the three sensory patches are segregated and Lmx1a-GFP is present around the utricle and in between the two cristae (star in **a**). High magnification surface **(b-e)** and transverse reconstructions (**f-g**) of the interface domain (brackets). The boundary of Lmx1a-GFP expression is relatively straight (arrowheads) and Sox2 expression is reduced in the cells immediately adjacent to it and facing the utricle (stars in **c**). A basal constriction is visible at the border of the Lmx1a-GFP expression domain (arrow in **f-g**). In the E15 *Lmx1a* $^{GFP/GFP}$ inner ear (**h-n**) GFP expression is reduced at the interface of the UT and the fused AC and LC (stars in **h**). At higher magnification (**i-n**), note the presence of GFP-negative cells in between GFP-positive cells at the interface domain (arrows in **i-l**) but a domain of reduced Sox2 expression (stars in **j**) is maintained between the fused cristae and the UT. (**m-n**) In orthogonal reconstructions, note the absence of a basal constriction at the border of the cristae domain. Black arrows on the side of images e and l indicate the corresponding position of z-reconstructions in f-g and m-n respectively.

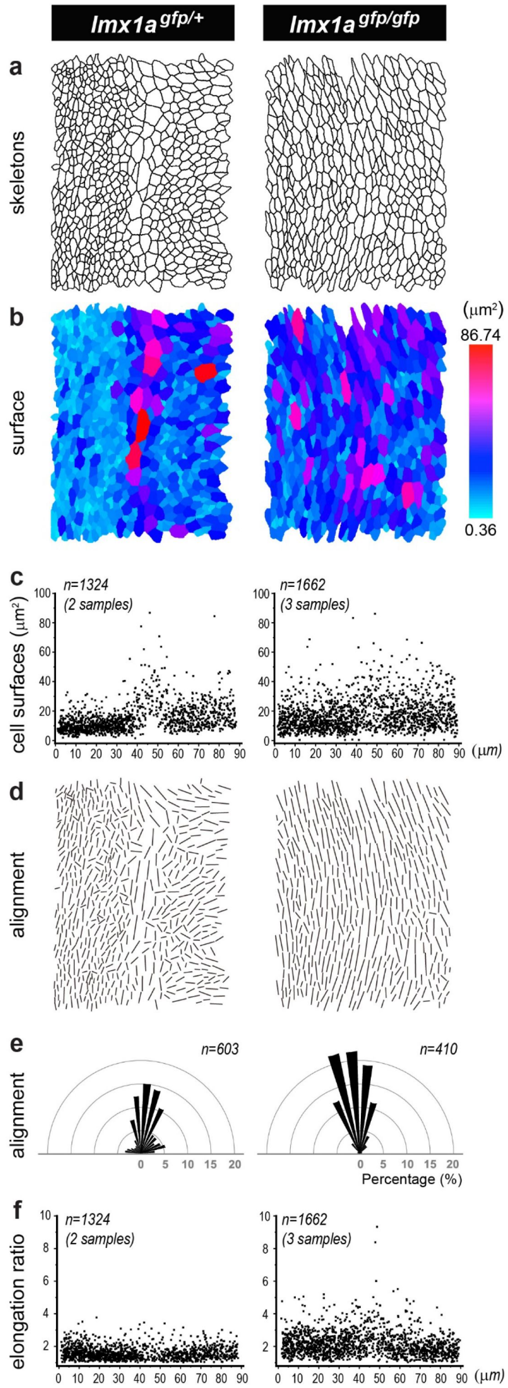


Figure 5 – Figure Supplementary 1.

Quantitative analysis of the boundary domain in *Lmx1a*$^{GFP/+}$ (left column) and *Lmx1a*$^{GFP/GFP}$ (right column) E15 mouse inner ear.

a) Representative cell skeletons generated from phalloidin stained preparations. In all images, the interface domain is in the middle, the utricle on the left, and the LC/AC domain on the right. **b-c**) Colour scale representation (from skeletons shown in **a**) and pooled scatter plots (from 2 *Lmx1a*$^{GFP/+}$and 3 *Lmx1a*$^{GFP/GFP}$ samples) of the surface cell areas. Note the more uniform distribution of the large cells on both sides of the interface domain in the Lmx1a-null inner ear. **d-e**) Visualization of the long axis (d) and angular distribution (e) of the cell skeletons shown in **a**). **f)** Pooled elongation ratio of the cells from 2 *Lmx1a*$^{GFP/+}$and 3 *Lmx1a*$^{GFP/GFP}$ samples. Note that cells are more aligned with one another and more elongated in the homozygous than in the heterozygous samples.

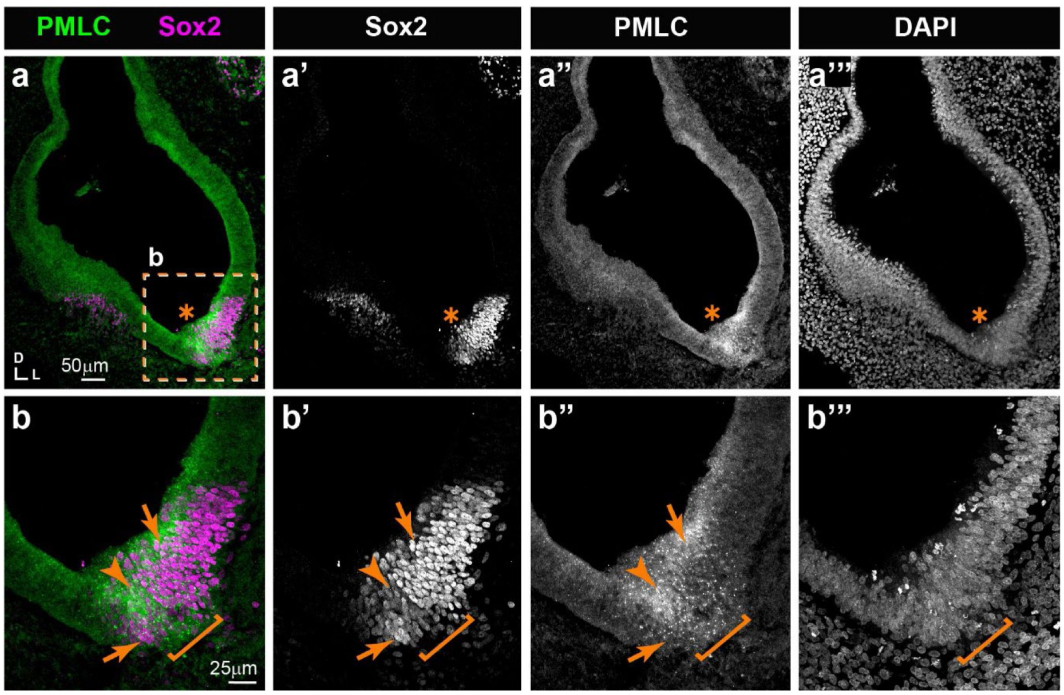


Figure 6

Expression pattern of phospho-myosin light chain (PMLC) in cross-sections of an E5 chicken inner ear.

a-a′′′) Punctate PMLC staining is present in the ventral part of the inner ear where a Sox2-positive prosensory domain (star) is localised. **b-b′′′)** High magnification views of the boxed area in **a.** The interface between the strong and weak Sox2 staining domains (arrows), typically corresponding to the boundary domain (bracket), exhibits an noticeable enrichment of PMLC staining at the surface and basal aspects of the epithelium (arrowheads).

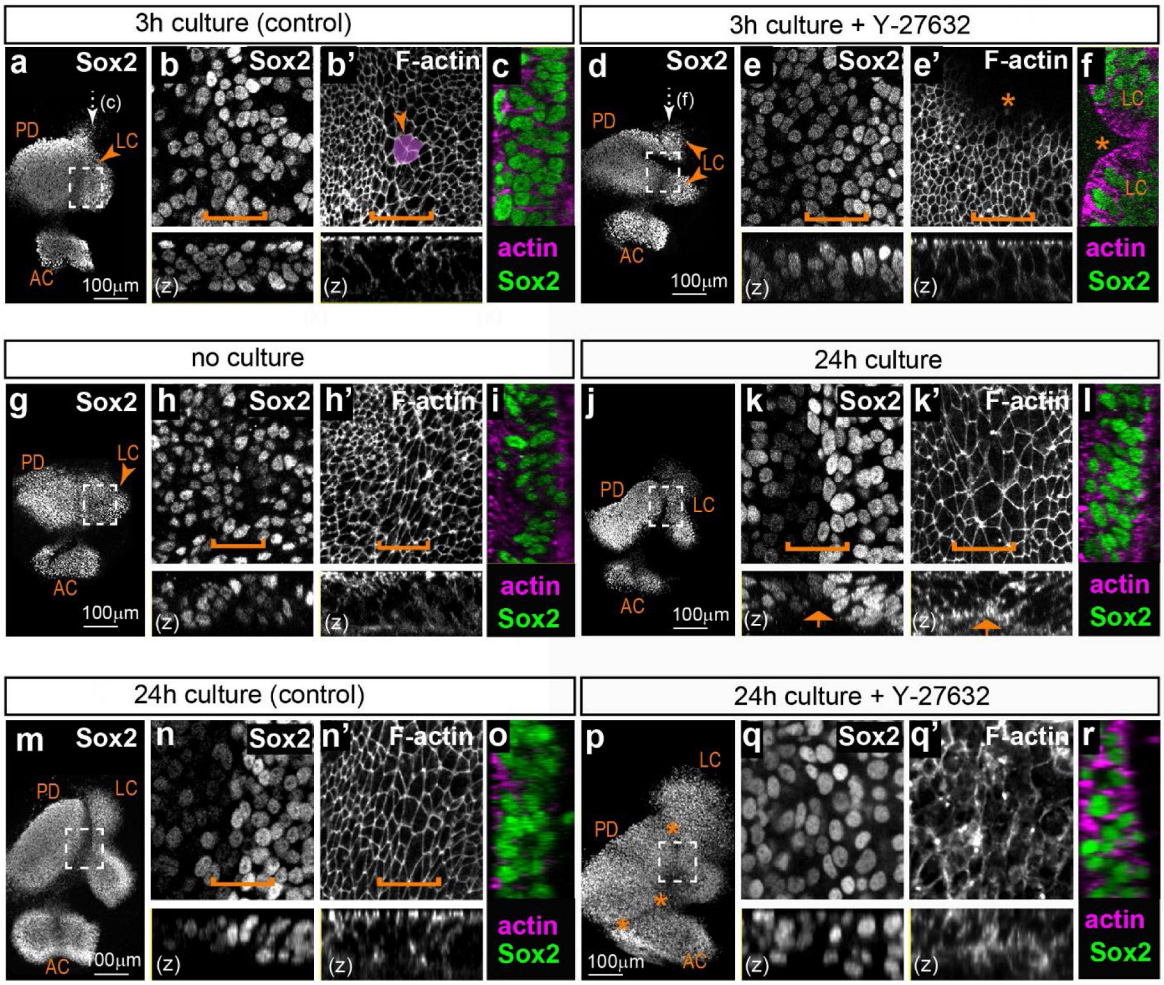


Figure 7

Effects of the ROCK inhibitor Y-27632 on the segregation of the lateral crista in organotypic cultures of embryonic chicken otocysts.

a-f) Surface and orthogonal views (**z, c, f**) of organotypic cultures of stage HH26 otocysts maintained for 3 hours in control medium (**a-c**) or medium supplemented with 20µm Y-27632 (**d-f**). Panels **b-c** and **e-f** are high magnification views of the boxed area in respectively **a** and **d**; the position of the large cell domain is indicated by a bracket and an example of multicellular rosette is highlighted (arrowhead in **b'**). Treatment with Y-27632 induces a loss of multicellular rosettes and abnormal folding of the epithelium (stars in **e'-f**). **g-l)** Surface and orthogonal views (**z, i, l**) of a pair of otocysts from the same HH26 embryo; one otocyst was fixed immediately (**g-i**) and the other maintained for 24 hours in culture (**j-l**). Panels **h-i** and **k-l** are high magnification views of the boxed area in respectively **g** and **j**; the position of the large cell domain is indicated by a bracket. After 24 hours, the lateral crista has progressed in its segregation from the pan-sensory domain. There is a clear reduction in Sox2 expression levels at the edge of the prospective LC and a basal constriction is noticeable (arrows in **k-k'**). **m-r)** Surface and orthogonal views (**z, o, r**) of a pair of otocysts from the same HH26 embryo; one otocyst was cultured 24 hours in control medium (**m-o**) and the other maintained for 24 hours in 20µm Y-27632 (**p-r**). Panels **n-o** and **q-r** are high magnification views of the boxed area in respectively **m** and **p**. Treatment with Y-27632 induces a loss of the typical epithelial cell morphology and large cell domain (compare **n'** and **q'**) and a disruption of sensory organ segregation (stars in **p**). Abbreviations: LC=lateral crista; AC=anterior crista; PD=pan-sensory domain.

nuclei of the lateral crista and utricle was not as well defined as in freshly fixed tissue, suggesting a partial disruption of tissue organisation *in vitro*. The boundary domain was still present in samples treated for 3h with 20µM Y-27632 (n=3; **Figure 7d-f**), but there was no basal constriction. Multicellular rosettes were completely absent and we noted in all samples some folding and invaginations of the epithelium, possibly caused by the inhibition of cortical actomyosin contractility by Y-27632. Nevertheless, the interface between the LC and the utricle was still visible, indicating that a short blockade of actomyosin contractility does not result in extensive cell mixing at the interface of the two organs. Furthermore, a quantitative analysis of the effects of Y-27632 treatment (**Figure 7 -Figure Supplementary 1a-f**) did not show significant changes in the size and alignment of surface cells in treated samples compared to controls.

Next, we examined the consequences of longer Y-27632 treatments. We first performed some experiments to determine whether the initial stages of LC segregation can occur *ex vivo*. We compared LC size and segregation in 15 pairs of otocysts, with each pair from the same embryo (HH26 – stage 1 or 2 of LC segregation) composed of an otocyst immediately fixed after dissection and the other one maintained in culture for 24h before fixation (**Figure 7g-l**). All of the cultured LC showed some progression in LC formation compared to their contralateral control, with a clear boundary between weakly and strongly Sox2-expressing cells in 12 out of 15 samples and a basal constriction present in 14 out of 15 samples. The size of the LC was greater in cultured samples, suggesting growth and survival of the prosensory cells. However, the distance between the LC and the pan-sensory domain ranged from 15.82-49.82 µm only, which was reduced compared to what would normally be observed *in ovo* in a stage HH27 otocyst (n=9; distance = 46.3-179.62 µm; see also Mann et al., 2017). These results show that the initial stages of specification and segregation of the LC do occur *in vitro*, albeit at a slower pace than *in ovo*.

We next investigated cell morphology and organisation in contralateral HH26 otocysts cultured for 24h in either control medium or 20µm Y-27631 (**Figure 7m-r**). Only three out of 6 Y-27631-treated samples could be analysed, due to the extreme fragility of the tissue which made impossible the dissection and mounting of the samples. Compared to their contralateral controls, the Y-27631-treated samples exhibited an extensive fusion and expansion of both the AC and LC domains, which were nevertheless present and distinguishable from the pan-sensory domain. Signs of severe epithelial disorganisation included the frequent absence of F-actin staining at the apical surface, suggesting a disassembly of the adherens junction complexes, and Sox2-positive nuclei located very close to the surface. The interface domain did not exhibit the typical down-regulation in Sox2 expression (**Figure 7p-q**) and had a statistically significant reduction in the number of large cells compared to controls (**Figure 7 – Figure Supplementary 1a-d**).

These results indicate that ROCK activity might be dispensable for the specification and growth of the LC, but it is absolutely required for the maintenance of its epithelial character and its segregation from the pan-sensory domain.

Perturbation of ROCK activity in ovo disrupts the organization of the boundary domain

In light of the very severe defects in epithelial morphology caused by long-term Y-27631 treatment and the limited extent of cristae segregation *in vitro*, we turned to a genetic approach to interfere with ROCK-dependent actomyosin contractility*in ovo* in the developing otocyst. We used *in ovo* electroporation (EP) of the otic cup to overexpress in the otocyst a truncated dominant-negative form of ROCK fused to GFP (RIIC1-GFP), able to bind to its upstream regulator Shroom3, but lacking the kinase domain necessary for Myosin II regulation (Nishimura and Takeichi, 2008). Pilot experiments on HH23 samples transfected with a transient expression vector confirmed that RIIC1-GFP localizes at the cell membrane (**Figure 8a**). The analysis of surface cell morphologies with Epitool (**Figure 8b-d**) showed that RIIC1-GFP causes a mild (approximately 23%) but

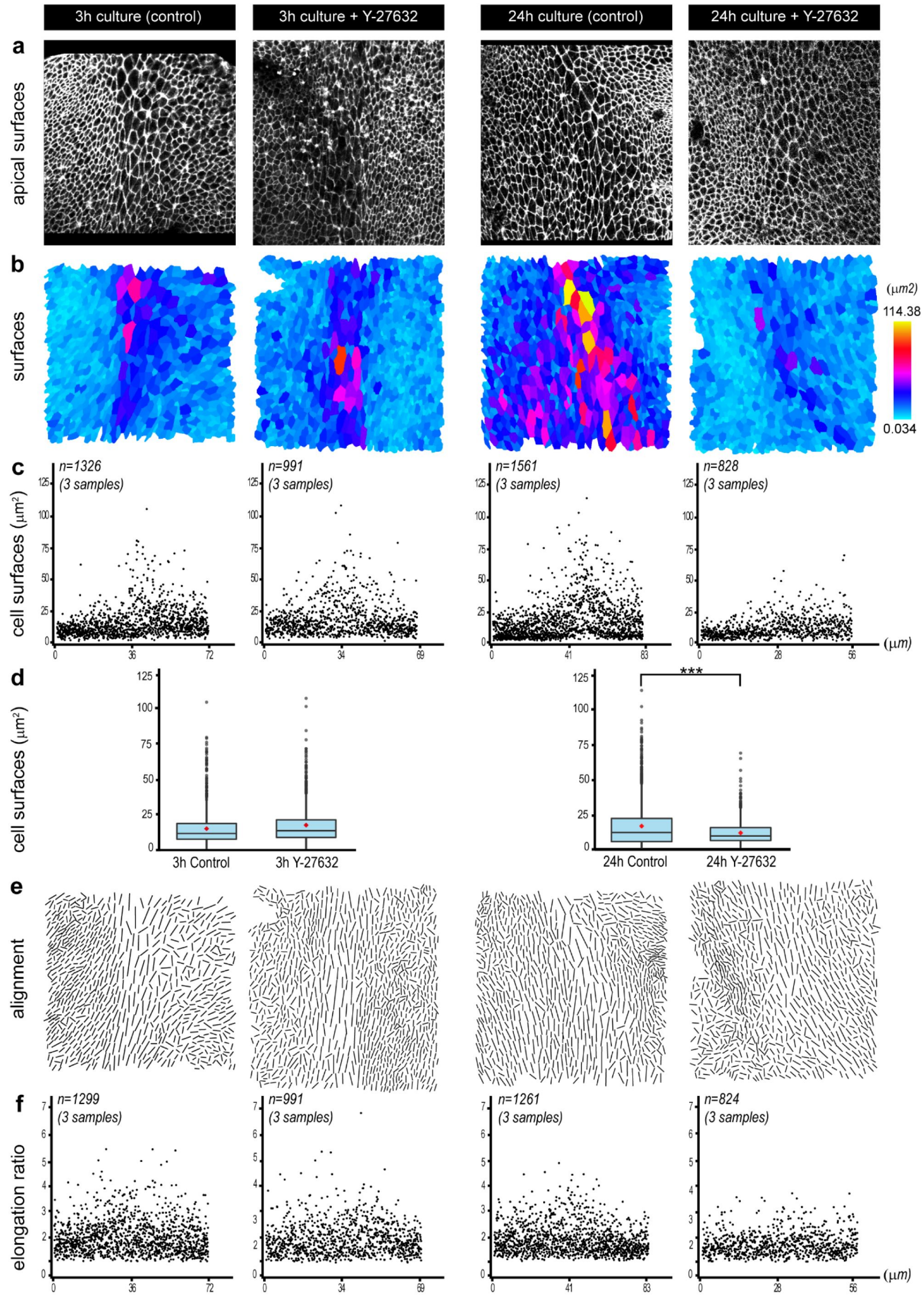


Figure 7 – Figure Supplementary 1.

Quantification of the effects of ROCK inhibition on cell morphology and organisation during segregation of the lateral crista.

a) Representative images of phalloidin-stained apical surfaces of epithelial cells from organotypic cultures maintained for 3 and 24 hours in control medium, or medium supplemented with 20μm Y-27632. In all images, the interface domain is in the middle and the lateral crista on the right. **b)** Colour scale representation (from skeletons shown in **a**). **c)** pooled scatter plots (3 samples per condition) of the cell surface areas and **d)** pooled box plots (bar=median; red dot=mean) of the surface areas of cells located within a +/-standard deviation from the location of the largest cells (the presumptive boundary cells) of each sample; there is a statistically significant difference between 24h control and Y-27632 treated samples (Mann Whitney test W=2801, p=2.2e^{-16}). **e-f)** Visualization of the long axis (from skeletons shown in **b**) and pooled elongation ratio of the epithelial cells.

statistically significant ($p<0.01$, Mann Whitney test U=10622.5) increase in the average apical surface area of transfected otic cells (18.22µm^2; SD=11.04) compared to untransfected cells (14.84 µm^2; SD=8.38), consistent with an inhibition of cortical actomyosin contractility. In comparison, cells transfected with a control RCAS-GFP construct (**Figure 8e**) do not have significantly enlarged apical surface area (p = 0.56, Mann Whiteny test U=77504).Next, we used the RCAS retroviral system (Hughes, 2004) to investigate the effects of RIIC1-GFP on cristae segregation, since EP of a replication-competent retrovirus proviral DNA leads to a sustained and widespread transgene expression (Freeman et al., 2012). The right otic cups of E2 (HH13-14) embryos were electroporated with either the RCAS-GFP (control) or the RCAS-RIIC1-GFP proviral DNA and the embryos were incubated for 3 to 5 days post-EP. In controls (n=8; **Figure 8g-h**’’’), the LC was present in 6 out of 8 samples and separated from the utricle by distances ranging from 55 to 155 µm. We hypothesize that the absence of LC in two of the controls may be due to abnormal or delayed development of the inner ear after EP. The pattern of GFP expression was very variable among samples, but in general it encompassed sensory and non-sensory regions. However, in two samples, we noted a very marked difference in GFP fluorescence levels on each side of the boundary of the LC (**Figure 8h**’). Although several factors could potentially explain this observation (e.g. different rate of GFP production or degradation in different cell populations), it fits with the idea that there is limited cell intermixing at this particular location.

In the RCAS-RIIC1-GFP electroporated samples (n=6; **Figure 8i-l**’’’), the LC was detected in all cases but it was either connected to the utricle (n=2/6), or migrated at a smaller distance than in controls (n=4/6; **Figure 8f**), suggesting an impaired segregation. The examination of the putative LC boundary domain at high magnification showed that as previously observed with the short-term overexpression experiments, RIIC1-GFP fluorescence was enriched at the cell surface and baso-lateral membranes (**Figure 8j**’ and **l**’).

In 4 out of 6 analysed samples, clear defects in the alignment of Sox2-positive cells were observed at the border of the prospective LC in regions with high levels of RCII-GFP expression, suggesting potential intermixing between cells of the LC and those of prospective non-sensory domains. Although it was not always possible to visualize it clearly, a basal constriction appeared to be present, including in domains with high levels of RCII-GFP expression (**Figure 8j’-j’’’**). Strikingly, in one sample with a mosaic pattern of RCII-GFP expression in the boundary domain, the untransfected cells were the only ones exhibiting a clear alignment of their apical borders reminiscent of what was observed in controls (**Figure 8l’-l’’’**).

We conclude that ROCK-dependent actomyosin contractility is required for the coordination of cortical cell tension at the boundary between the LC and pan-sensory domain and for normal sensory organ segregation. However, additional mechanisms are likely to limit the intermixing of prosensory cells at the boundary since the separation of Sox2-high and Sox2-low cells was still visible in these experiments.

The boundary domain is highly proliferative

One important function of actomyosin-dependent tissue boundaries is to limit the disruptive impact of cell proliferation on the spatial organisation of cells during tissue growth and morphogenesis (reviewed in Monier et al., 2010). This prompted us to examine the spatial pattern of cell proliferation during LC segregation. To detect S-phase cells, we administered *in ovo* a short (2 hour) pulse of the thymidine analogue EdU to stage HH23-25 embryos before collecting inner ear tissue and process it for Click-iT EdU reaction and Sox2 immunostaining. Numerous EdU-positive cells were present throughout the boundary domain at both early (stage 1, **Figure 9a-b**; n=7) and advanced (stage 2-3, **Figure 9c-d**; n=7) segregation stages, and these always included some of the cells located at the interface of the prospective LC and pan-sensory domain. This was confirmed by the seemingly random distribution of mitotic figures (visualized by DAPI staining)

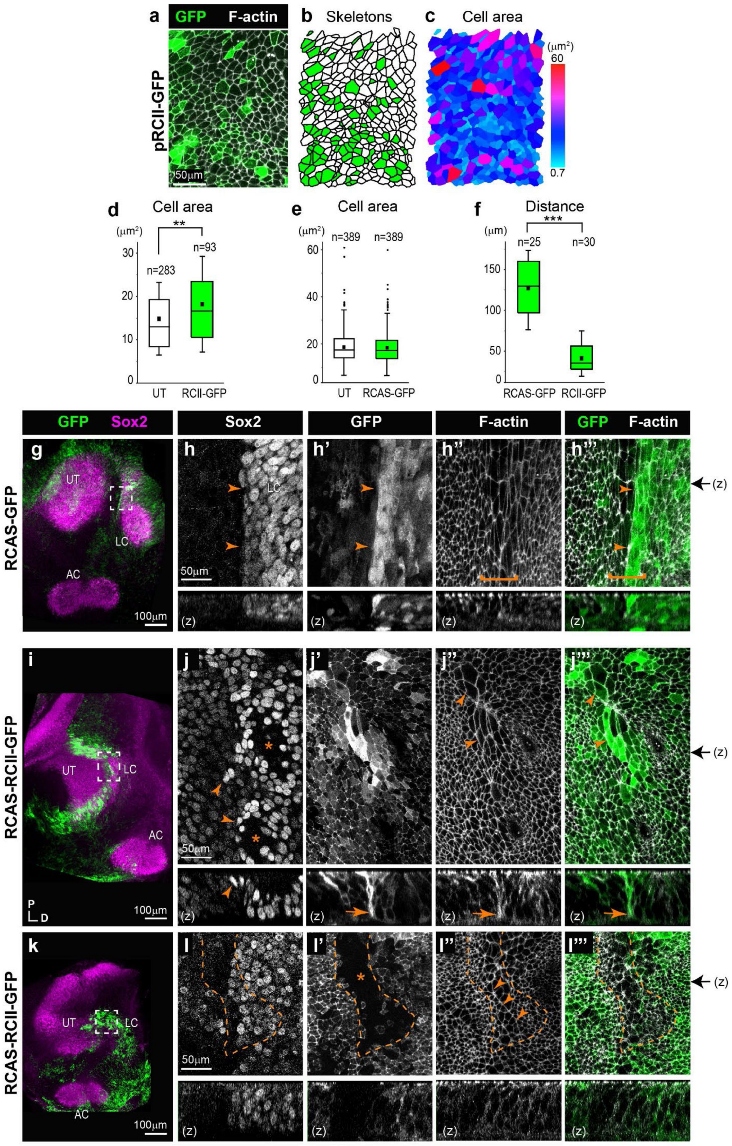


Figure 8

Overexpression of a dominant-negative form of ROCK in the developing chicken inner ear disrupts sensory organ segregation.

a) surface view of otic epithelial cells 24h after in ovo electroporation with the pRCII-GFP expression plasmid. **b-e)** analysis of cell skeletons of transfected (in green) versus untransfected (UT) cells indicate that overexpression of the RCII-GFP fusion protein induces an increase in apical cell surface area compared to control electroporation with RCAS-GFP (e). **f**) quantification of the distance between the LC and UT in ears transfected with pRCII-GFP expression plasmid and control RCAS-GFP. **g-h′′′)** surface views of an E6 control otocyst electroporated at E2 with an RCAS-GFP proviral DNA. In this sample, the limit of Sox2 expression (arrowheads in h) matches that of strong GFP fluorescence (arrowheads in h′ and h′′′) and the presumptive tissue boundary in the middle of the large cell domain (bracket in h′′ and h′′′) separating the lateral crista (LC) from the utricle (UT). **i-l′′′)** surface views of two E6 samples electroporated at E2 with RCAS-RCII-GFP and showing abnormal segregation of the LC. In the first example, the interface domain contains poorly aligned Sox2-expressing cells (arrowheads in j). A basal constriction is still visible (arrows in **j′-j′′′**, transverse reconstructions) but the large cells of the interface domain are not properly aligned (arrowheads in **j″** and **j′′′**). In the second example (**k-l′′′**), a mosaic pattern of RCII-GFP fluorescence is observed in the interface domain. Cells with low levels of RCII-GFP fluorescence (star in **l′**) exhibit a more regular alignment of their apical borders (arrowheads in **l′′**) than transfected cells. The black arrows on the right side of images h′′′, j′′′, and l′′′ indicate the corresponding position of z-reconstructions.

within the boundary domain of fixed samples (n=7 for each stage; **Figure 9b, d** and **e**). We conclude that the boundary at the interface of the LC and pan-sensory domains contains mitotic cells throughout the segregation stages.

Discussion

We discovered a specialized actomyosin-dependent boundary domain at the interface of segregating sensory organs in the developing inner ear. We propose that this boundary restricts intermixing between Lmx1a-positive and negative cells during the segregation of adjacent sensory organs and act as a signalling centre guiding the differentiation of non-sensory territories. Our findings provide strong support to the 'compartment boundary' model for inner ear morphogenesis (Brigande et al., 2000b; Fekete, 1996; Fekete and Wu, 2002) and suggest that an evolutionary conserved module for tissue boundary formation has been co-opted in the inner ear for separating its distinct sensory organs.

A tissue boundary forms during the segregation of inner ear sensory organs

The morphogenesis of complex tissues depends on the harmonious coordination of cell proliferation, cell specification, and tissue patterning events. Tissue boundaries play an important role in this process by preventing, at specific locations, the intermingling of cells destined to form distinct anatomical structures. Besides this 'physical' role, tissue boundaries can act as signalling centres or 'organizers', regulating the proliferation and differentiation of adjacent cells through various intercellular signalling pathways. Tissue boundaries were discovered (Garcia-Bellido et al., 1973) and most studied in the imaginal wing disc of Drosophila, where they establish lineage-restricted compartments along the dorso-ventral and antero-posterior axis (Wang and Dahmann, 2020). In vertebrates, they are present in the nervous system (Addison and Wilkinson, 2016; Langenberg and Brand, 2005), in the ectoderm of the limb bud (Altabef et al., 1997), or in the gut (Smith and Tabin, 2000). Their formation typically involves three consecutive steps (Addison and Wilkinson, 2016; Dahmann et al., 2011; Wang and Dahmann, 2020). Firstly, 'selector' genes (encoding transcription factors) assign a specific identity to different populations of cells. Secondly, these cells interact through signalling molecules and sort themselves into adjacent domains according to their genetic identity through adhesive or repulsive interactions. Finally, actomyosin-dependent processes increase cell tension at the interface of adjacent compartments to create and maintain a stable 'fence' preventing cell mixing (Major and Irvine, 2006; Monier et al., 2011, 2010).

The morphogenesis of the inner ear has long been proposed to depend on the formation of lineage-restricted embryonic compartments analogous to those present in the fly wing disc. In fact, some genes have sharply defined domains of expression in the otic vesicle before any sign of morphological differentiation and their absence prevents the formation of specific structures of the adult inner ear, suggesting that these may act as selector genes for a particular otic fate (Brigande et al., 2000b; Fekete, 1996; Fekete and Wu, 2002). Fate map and lineage studies have also suggested the existence of lineage-restricted boundaries in the dorsal part of the chicken otic vesicle, which gives rise to the endolymphatic duct and sac, although their exact location and cellular features remain unknown (Brigande et al., 2000a; Kil and Collazo, 2002; Sánchez-Guardado et al., 2014). In this study, we provide strong evidence that the segregation of the anterior and lateral cristae from the prospective utricle is associated to the formation of a tissue boundary. It is composed of a subset of prosensory cells which progressively enlarge, elongate, and align their apical cell borders at the precise junction between Lmx1a-positive and negative cells (**Figure 10**). In the absence of lineage-tracing or live-imaging experiments, we do not have definitive proof that these cells generate a strict lineage-restricted boundary. However, their position (at the interface of two distinct "genetic compartments") and cellular features (basal

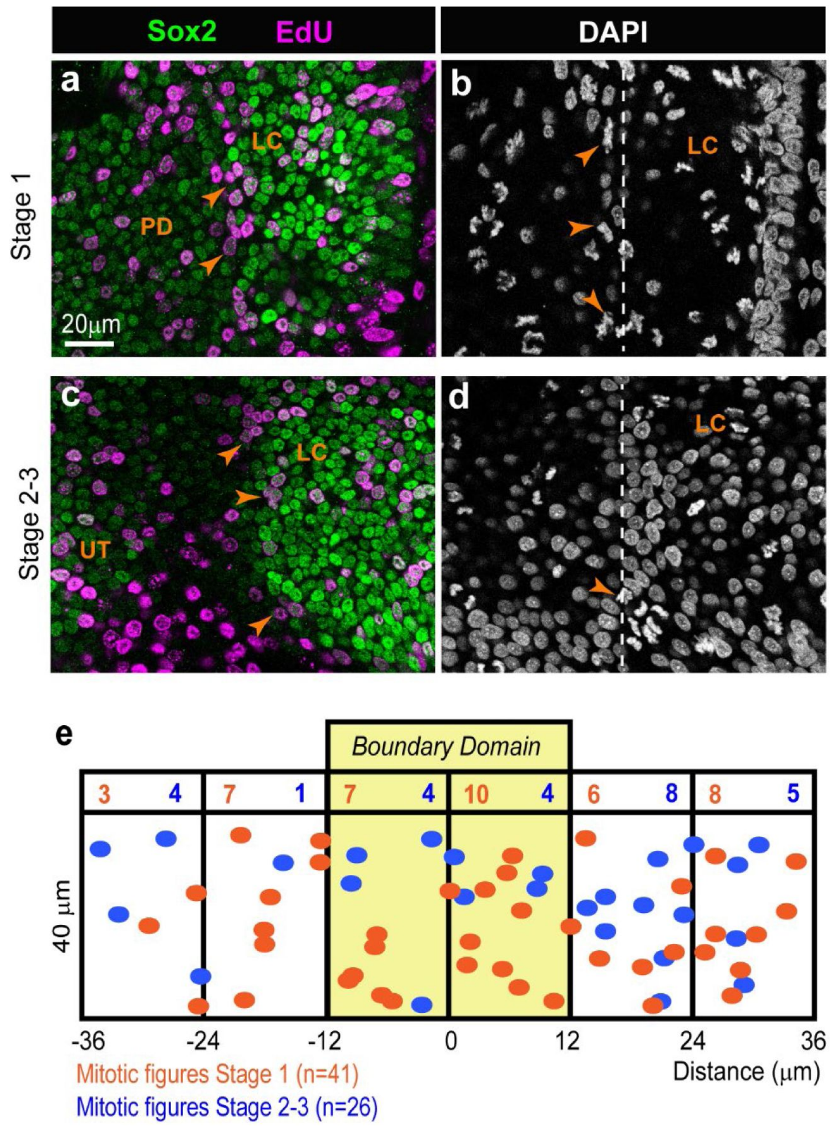


Figure 9.

Analysis of cell proliferation during the segregation of the lateral crista in the chicken otocyst.

whole-mount preparations of samples collected one hour after an EdU pulse in ovo at stage 1 (**a-b**) or stage 2-3 (**c-d**) of segregation of the LC, immunostained for EdU, Sox2 and labelled with DAPI. Numerous EdU-positive cells (**arrows in a-c**) and mitotic figures (**arrows in b-d**) are present at the interface between the LC and PD (**dotted line in b-d**) at both stages. **e)** Spatial distribution of the mitotic figures within a 36×72μm region centred on the middle of the boundary domain, for early and late stages of segregation (n=7 embryos for each stage).

constriction, basal multicellular actin-cable-like structure, and alignment of their apical cell borders) are consistent with this idea. Furthermore, a lineage-restricted boundary could explain why we observed, in some of our RCAS-infection experiments, an uneven distribution of GFP-expressing cells on either side of the basal constriction separating the lateral crista from the utricle. In a previous study, we argued that the changing patterns of Lmx1a and Sox2 expression in the embryonic inner ear suggested some form of dynamic competition for the adoption of the sensory versus non-sensory fates at the lateral border of sensory organs, as opposed to the existence of a strict lineage boundary (Mann et al., 2017). Our new results, which focused on earlier stages of sensory organ formation, call for a revision of this conclusion: the early prosensory cells are indeed labile, but their initial separation into distinct pools of sensory progenitors (for the cristae and the utricle at least) appears to be associated to the formation of a transient lineage-restricted tissue boundary.

Which mechanisms establish and maintain the boundary domain between segregating sensory organs? In a previous study, we showed that Lmx1a is essential for the specification of the non-sensory territories separating sensory organs and its overexpression in the chicken inner ear antagonizes prosensory specification, suggesting that it acts as a “selector” gene for the non-sensory fate (Mann et al., 2017). The present study confirms this hypothesis: in its absence (in *Lmx1a*$^{GFP/GFP}$embryos), cells of the Lmx1a-lineage commit to a prosensory fate, leading to a fusion between the lateral and anterior cristae. On the other hand, the fused cristae and the utricle are separated by a Sox2-negative domain, suggesting that the initial segregation of these two sensory domains proceeds independently from Lmx1a function. Nevertheless, the typical features (large cell domain and basal constriction) of the boundary domain separating the cristae from the utricle were absent in all but one of the *Lmx1a*-null samples examined. Altogether, these results suggest that Lmx1a plays a role in the maintenance or continued differentiation of the boundary domain, but is not required for its initial formation. Finally, the patterning abnormalities in *Lmx1a*$^{GFP/GFP}$samples occurred in both GFP-positive and negative territories, which points at some type of interaction between Lmx1a-expressing and non-expressing cells, and the possibility that the boundary domain is also a signalling centre influencing the differentiation of adjacent territories.

Given its conserved role in tissue boundaries, we hypothesized that actomyosin contractility could contribute to the formation or maintenance of the boundary domain. The *in vitro* results showed that short-term inhibition of ROCK-dependent actomyosin contractility with Y-27632 disrupts the morphology and patterning of boundary cells. Interestingly, the large cells were still present, but more elongated and more polarised than in control cultures. This suggests that global anisotropic forces within the developing otocyst contribute to a large extent to the elongated cellular shapes and long axis alignments observed at the interface of segregating organs. Nevertheless, the spatial separation of cells expressing either high or low levels of Sox2 was maintained. Longer-term treatments led to an apparent fusion of adjacent sensory territories, but this may have been caused by the complete loss of epithelial integrity observed in these experiments. On the other hand, the overexpression in the chicken inner ear of a dominant-negative form of ROCK, unable to interact with its upstream regulator Shroom3 (Nishimura and Takeichi, 2008), produced more informative results. Firstly, the LC failed to segregate normally from the pan-sensory domain, indicating a requirement for ROCK activity in this process. Secondly, the border of the LC was very irregular, with some intermingling between cells expressing high and low levels of Sox2. This would be expected if an actomyosin-dependent lineage boundary had been disrupted. Altogether, these results show that ROCK-dependent coordination of actomyosin contractility is crucial for the segregation of the sensory organs and the maintenance of a straight boundary between these, yet other mechanisms are likely to contribute to the initial sorting of the Lmx1a-expressing and non-expressing cells. One strong candidate is the basal constriction and actin-cable-like structure at the edge of the cristae, which resembles the one described at the midbrain-hindbrain boundary (Gutzman et al., 2018, 2008) and could contribute to the physical separation of sensory progenitors. It has been proposed that a zone of non-proliferating cells could impede cell

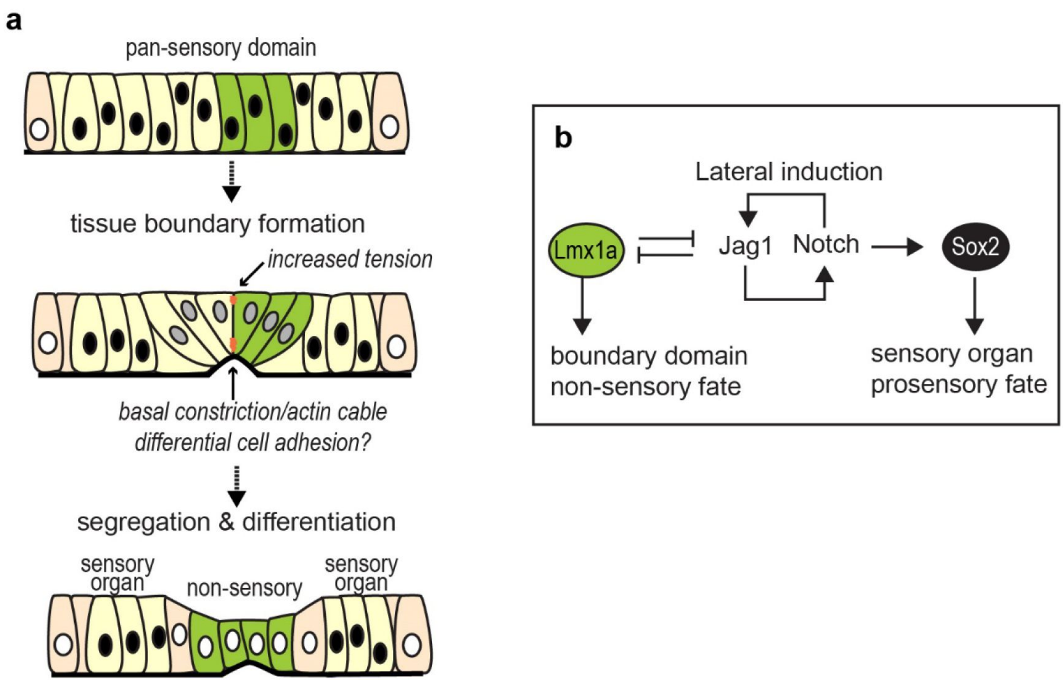


Figure 10.

A schematic representation of sensory organ segregation.

a) The up-regulation of Lmx1a expression within the pan-sensory domain coincides with the formation of a specialised boundary domain composed of cells with enlarged cell surfaces. At the interface of Lmx1a-expressing and non-expressing cells, cells align their borders apically and form multicellular basal constriction s and actin-cable-like structures at the base of the epithelium. We propose that it is a transient lineage-restricted boundary, dependent on actomyosin contractility, which separates adjacent pools of sensory progenitors. As the spatial segregation proceeds, the Lmx1a-expressing cells give rise to non-sensory cells separating sensory organs. **b)** Hypothetical regulatory gene network regulating sensory organ segregation. Notch-dependent lateral induction maintains Sox2 expression and promotes the prosensory fate. Lmx1a antagonizes Notch activity and promotes adoption of a non-sensory fate; it is also required for the proper formation of the boundary domain between the cristae and utricle.

movements between adjacent compartments at the DV wing disc boundary (Blair, 1993; see however O'Brochta and Bryant, 1985 for a different view) or between rhombomeres in the hindbrain (Guthrie et al., 1991). However, the results of our EdU-incorporation assays show extensive proliferation of the cells composing the boundary domain at early and late stages of segregation, arguing against this hypothesis. Other mechanisms involved in boundary formation in the hindbrain and insect compartments include differential cell adhesion or repulsive interactions mediated by eph-ephrin signalling (Dahmann et al., 2011; Wilkinson, 2018). It is also possible that the repositioning of cells through medial intercalation could contribute to the straightening of the boundary as well as the widening of the non-sensory territories in between sensory patches. Further studies will be needed to determine which of these mechanisms, if any, contributes to sensory organ segregation in the inner ear.

Deep homologies between sensory organ segregation and drosophila wing disc compartmentalization: co-option of a tissue boundary module?

An increase in cortical cell tension or a multicellular actin-cable have been described along many tissue boundaries (Fagotto, 2014; Wang and Dahmann, 2020): the dorso-ventral (Major and Irvine, 2006, 2005) and antero-posterior (Landsberg et al., 2009) boundaries of the wing disc, the parasegments (Monier et al., 2010) and salivary gland placode (Röper, 2012) boundaries in drosophila and at the interface of rhombomeres of the hindbrain (Calzolari et al., 2014) in vertebrates. Hence, the implication of actomyosin-dependent processes to prevent cell intermixing is a conserved feature of tissue boundaries, which is also present in the inner ear. Perhaps more surprisingly, the core elements of the gene regulatory network regulating sensory organ segregation in the inner ear and the dorso-ventral (DV) compartmentalization of the drosophila wing disc are remarkably similar (Daudet and Żak, 2020). In fact, Notch signalling acts by lateral induction in both tissues: to maintain prosensory character in the inner ear, or to enhance Notch activity at the DV boundary in the wing disc. We have argued that *Lmx1a* and *Apterous*, which are orthologous genes, play a selector-like function in both tissues, specifying the identity of dorsal cells in the wing disc and that of the non-sensory cells separating the vestibular organs in the inner ear. In both tissues, the interactions between Apterous/Lmx1a and Notch signalling have a comparable outcome: a reduction in the levels of Notch activity within Apterous/Lmx1a expressing cells. In the wing disc, Apterous drives the expression of Fringe, a glycosyl-transferase that makes the Notch receptor less sensitive to the Serrate ligand in dorsal cells, potentiating signalling through Delta, expressed by ventral cells (de Celis et al., 1996; de Celis and Bray, 1997; Doherty et al., 1996; Irvine and Wieschaus, 1994, 1994; Panin et al., 1997; Rauskolb et al., 1999). We do not know yet if any of the vertebrate Fringe orthologues contribute to the regulation of Notch activity at the border of inner ear vestibular organs. However, it has been shown that Lfng and Mfng are expressed at the medial border of the prosensory domain in the embryonic mouse cochlea, and *Lnfg;Mfng* double mutants exhibit patterning defects consistent with an abnormal positioning of the medial border of the organ of Corti (Basch et al., 2016b). Further studies should investigate the roles of Fringe proteins during vestibular organ segregation. By extension, it would be worth investigating the potential role of Wnt signalling in this process, since one of the important consequences of Notch activity at the dorso-ventral wing disc boundary is to induce the expression of Wingless, which then regulates the growth and patterning of the wing (de Celis et al., 1996; Diaz-Benjumea and Cohen, 1995; Rulifson and Blair, 1995). It is worth noting that a similar genetic circuit including Notch signalling (Fringe, Jagged) and Lmx1 is deployed in vertebrate limbs to position the apical ectodermal ridge, the domain that act as a growth zone and lineage boundary between the dorsal and the ventral ectoderm of the limb (Altabef et al., 1997). This 'deep homology', defined as "the sharing of the genetic regulatory apparatus that is used to build morphologically or phylogenetically disparate animal features" (Shubin et al., 2009), suggests that Notch, Lmx1 and actomyosin contractility are part of an ancestral module for tissue boundary formation that has been co-opted and adapted for the patterning of a great variety of tissues.

Within the inner ear itself, it is possible that modifications of this core module are responsible for some of the interspecies differences in sensory organ numbers and morphologies. It is thought that the ancestral inner ear resembled that of jawless vertebrates (the lamprey and the hagfish), with only two cristae (anterior and posterior) and a single ventral macula (Hammond and Whitfield, 2006; Higuchi et al., 2019; Jørgensen et al., 1998; Lowenstein et al., 1968). Jawed vertebrates harbour three semi-circular canals and cristae as well as additional sensory organs in the vestibular and ventral part of the inner ear (Retzius, 1884, 1881), enabling in particular the detection of air-borne sound in land vertebrates. Differences in inner ear sensory organ numbers, size and morphologies are common place among vertebrates and it is possible that some of these originate from changes in the genetic networks controlling sensory organ segregation. In the present study, we found at least a clear indication of interspecies variations in the segregation of the cristae: in the chicken inner ear, the anterior crista segregates before the lateral one, whilst in the mouse the segregation of the two cristae occurs simultaneously. Furthermore, the chicken lateral crista is formed by two Sox2-positive lobes, one of which turns into a non-sensory territory. The fact that the specification of prosensory cells and their segregation is regulated by signalling pathways with well-known spatial patterning activities (FGF, Wnt and Notch signalling) may have offered ample opportunities for “sensory organ experimentation”, facilitating in the course of vertebrate evolution the emergence of additional organs with new functionalities, or on the contrary the suppression of redundant ones.

Material and methods

Animals

Fertilised White Leghorn chicken (*Gallus gallus*) eggs were obtained from Henry Stewart UK and incubated at 37.8°C for the designated times. Embryonic stages refer to embryonic days (E), with E1 corresponding to 24 hours of incubation or to Hamburger and Hamilton stages (Hamburger and Hamilton, 1951). Embryos older than E5 were sacrificed by decapitation.

Lmx1a^{GFP} knock-in mice (Griesel et al., 2011) were maintained on a C57BL/6 mixed background at the UCL Ear Institute animal facility and bred to generate Lmx1a$^{GFP/GFP}$ (homozygous/Lmx1a null) and Lmx1a$^{GFP/+}$ (heterozygous) mice.

All animal procedures were carried out in accord with United Kingdom legislation outlined in the Animals (Scientific Procedures) Act 1986 and approved by University College London local Ethics Committee.

In ovo electroporation

Electroporation of the otic placode/cup of E2 chick embryos (stage HH 10–14) was performed using a BTX ECM 830 Electro Square Porator as previously described (Freeman et al., 2012). The RIIC2-GFP plasmid was obtained from Nishimura, T. & Takeichi (Nishimura and Takeichi, 2008) and the insert subcloned into RCAS using conventional methods. All plasmids used for *in ovo* electroporation were purified using the 16 Qiagen Plasmid Plus Midi Kit (Qiagen). The total concentration of DNA ranged for each set of experiments between 0.5–1µg/µl. The electroporated inner ears were collected from the embryos between E5-7 and processed for whole-mount immunohistochemistry.

Immunohistochemistry

Entire chicken embryos (E3-E4) or their heads (>E5) were collected, fixed for 1.5-2 hours in 0.1 M phosphate buffered saline (PBS) with 4% paraformaldehyde and processed for whole-mount immunostaining as described previously (Mann et al., 2017). In brief, the inner ear epithelium was dissected out from surrounding mesenchyme and incubated for 30 minutes in a blocking solution consisting of PBS with 0.3% Triton and 10% goat serum. The primary and secondary antibodies (diluted in PBS with 0.1% Triton, or PBT) were incubated overnight at 4C or for 2 hours at room temperature. Several rinses in PBT were done after antibody incubation. The following reagents were used: rabbit anti-Sox2 (Abcam, UK; 97959, 1:500); mouse IgG1 monoclonal anti-Sox2 (BD Biosciences, San Jose, CA; 561469, 1:500); mouse IgG1 anti-phospho-Myosin light chain 2 (Ser19) (Cell Signaling Technology, MA, 3675, 1:200), rat IgG1 anti-Laminin-β1 (Abcam, AB44941, 1:50); secondary goat anti-mouse IgG antibodies and phalloidin conjugated to Alexa dyes (1:1000, Thermo Fischer Scientific UK). At the end of the immunostaining protocol, the samples were further dissected under a Leica MZ16F stereomicroscope equipped with a fluorescence light source to visualize precisely the position of the Sox2-expressing sensory organs and facilitate their mounting in Vectashield (Vector labs, UK). A fine layer of vacuum grease was applied between the slide and coverslip to avoid excessive flattening of the tissue. Confocal stacks were acquired using a Zeiss LSM880 inverted confocal microscope or a Perkin Elmer spinning-disc confocal and further processed with ImageJ.

Analysis of cell morphologies

Phalloidin labelling was used to visualize and analyse apical cell surfaces. Confocal z-stacks of the surface of the samples were processed with Image J and exported into Epitool for automated cell segmentation (Heller et al., 2016). The cell skeletons were manually corrected using CellEdit plugin in the Icy software (de Chaumont et al., 2012). Colour scale of cell surface area and elongation ratio were generated based on the corrected cell skeleton using CellOverlay plugin in Icy. Quantitative data on cell morphologies were exported into Excel spreadsheet and scatter plots, radial histogram, box plots and statistics of the data were generated using OriginPro 2017 and R. For the quantification of cell surfaces in Y-27632 experiments (**Figure 7 supplementary 1**), a subset of cells located within a +/-standard deviation interval from the location of the largest cells of each sample was analysed.

Organotypic cultures

The anterior domain of the otocysts and its underlying mesenchyme were dissected from E5 (HH26) chicken embryos in room-temperature L-15 medium (Leibovitz). The dissected tissues were either fixed at t=0 or cultured in 150-300μL DMEM medium (Invitrogen) containing 1% HEPES and 20 μm Y-27632 (Sigma-Aldrich, cat # Y0503, lot # 103M4711V; dissolved in DMEM) in a tissue culture incubator (5% CO_2, 37 degree) for 3h or 24hr. For the 3h-long experiments, the tissues were placed in 96-well plates and free floating in DMEM medium. For the 24h-long experiments, the tissues were placed in 35mm Mattek dishes and coated with a small (approximately 20μl) drop of 0.4%-0.8% agarose (low gelling temperature, Sigma-Aldrich) in 0.1M PBS before adding DMEM medium. The cultured tissues were then fixed and processed for immunostaining as previously described.

Cell proliferation assays

Fertilized eggs were windowed at E4-5 (HH23-25) and a 200μl drop of a PBS solution containing 200μM of EdU (5-ethynyl-2’-deoxyuridine) was applied on top of the embryo. The eggs were sealed with tape and returned to incubation for 2 hours, then the embryos heads were collected and fixed for 30 min at room temperature in PBS with 4% Formaldehyde. Following several rinses in PBS and PBT, the inner ear tissue was dissected and processed for Click-iT EdU reaction (Thermofisher UK) according to the manufacturer’s protocol. At the end of the reaction, the tissue was further processed for Sox2 immunostaining as described above and labelled with DAPI before mounting. For the mapping of mitotic figures (identified by DAPI staining), a 72×36μm “region-of-interest” centred on the boundary between the lateral crista and pan-sensory domain was selected from a confocal stack acquired with a 40x objective. The position of each mitotic figure was recorded in x,y and the results plotted for different samples collected at stage 1 (n=7) or 2-3 (n=7).

Measurement of the distance between segregating sensory organs

The distance between the lateral crista and the utricle was measured in maximum intensity projections of z-stacks using ImageJ. The distance (μm) was measured in five different positions along the boundary domain in each analysed sample. Next, all the results were analysed in R using Two-sample Kolmogorov-Smirnov test and two-sided Mann-Whithey-Wilcoxon test.

Data availability

All data generated or analysed during this study will be available upon request.

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Reviewer #1 (Public review):

Summary:

This manuscript investigated the mechanism underlying boundary formation necessary for proper separation of vestibular sensory end organs. In both chick and mouse embryos, it was shown that a population of cells abutting the sensory (marked by high Sox2 expression) /nonsensory cell populations (marked by Lmx1a expression) undergo apical expansion, elongation, alignment and basal constriction to separate the lateral crista (LC) from the utricle. Using Lmx1a mouse mutant, organ cultures, pharmacological and viral-mediated Rock inhibition, it was demonstrated that the Lmx1a transcription factor and Rock-mediated actomyosin contractility is required for boundary formation and LC-utricle separation.

Strengths:

Overall, the morphometric analyses were done rigorously and revealed novel boundary cell behaviors. The requirement of Lmx1a and Rock activity in boundary formation was convincingly demonstrated.

Weaknesses:

However, the precise roles of Lmx1a and Rock in regulating cell behaviors during boundary formation were not clearly fleshed out. For example, phenotypic analysis of Lmx1a was rather cursory; it is unclear how Lmx1a, expressed in half of the boundary domain, control boundary cell behaviors and prevent cell mixing between Lmx1a+ and Lmx1a- compartments? Well-established mechanisms and molecules for boundary formation were not investigated (e.g. differential adhesion via cadherins, cell repulsion via ephrin-Eph signaling). Moreover, within the boundary domain, it is unclear whether apical multicellular rosettes and basal constrictions are drivers of boundary formation, as boundary can still form when these cell behaviors were inhibited. Involvement of other cell behaviors, such as directional cell intercalation and oriented cell division also warrant consideration. With these lingering questions, the mechanistic advance of the present study is modest.

Revision: The clarity of the text was improved. The open questions regarding the mechanisms were not experimentally addressed but discussed.

https://doi.org/10.7554/eLife.106305.2.sa2

Reviewer #3 (Public review):

Summary:

Lmx1a is an orthologue of apterous in flies, which is important for dorsal-ventral border formation in the wing disc. Previously, this research group has described the importance of the chicken Lmx1b in establishing the boundary between sensory and non-sensory domains in the chicken inner ear. Here, the authors described a series of cellular changes during border formation in the chicken inner ear, including alignment of cells at the apical border and concomitant constriction basally. The authors extended these observations to the mouse inner ear and showed that these morphological changes occurred at the border of Lmx1a positive and negative regions, and these changes failed to develop in Lmx1a mutants. Furthermore, the authors demonstrated that the ROCK-dependent actomyosin contractility is important for this border formation and blocking ROCK function affected epithelial basal constriction and border formation in both in vitro and in vivo systems.

Strengths:

The morphological changes described during border formation in the developing inner ear are interesting. Linking these changes to the function of Lmx1a and ROCK dependent actomyosin contractile function are provocative.

Weaknesses:

There are several outstanding issues that need to be clarified before one can pin the morphological changes observed being causal to border formation and that Lmx1a and ROCK are involved.

Comments on the latest version:

The revised manuscript has provided clarity of their results on some levels, but unfortunately, the basal restriction during border formation remains unclear and the study did not advance the understanding of role of Lmx1a in boundary formation. Overall comments are indicated below:

(1) The authors states in the rebuttal, "we do not think that ROCK activity is required for the formation or maintenance of the basal constriction at the interface of Lmx1a-expressing and non-expressing cells"
If the above is the sentiment of the authors, then the manuscript is not written to support this sentiment clearly, starting with this misleading sentence in the Abstract, "The boundary domain is absent in Lmx1a-deficient mice, which exhibit defects in sensory organ segregation, and is disrupted by the inhibition of ROCK-dependent actomyosin contractility."

(2) As acknowledged by the authors, the data as they currently stand could be explained by Lmx1a functioning in specifying the non-sensory fate and may not function directly in boundary formation. With this caveat in mind, the role of Lmx1a in boundary formation remains unclear.

(3) I feel like the word "orchestrate" in the title is an overstatement.

https://doi.org/10.7554/eLife.106305.2.sa1

Author response:

The following is the authors’ response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *This manuscript investigated the mechanism underlying boundary formation necessary for proper separation of vestibular sensory end organs. In both chick and mouse embryos, it was shown that a population of cells abutting the sensory (marked by high Sox2 expression) /nonsensory cell populations (marked by Lmx1a expression) undergo apical expansion, elongation, alignment and basal constriction to separate the lateral crista (LC) from the utricle. Using Lmx1a mouse mutant, organ cultures, pharmacological and viral-mediated Rock inhibition, it was demonstrated that the Lmx1a transcription factor and Rock-mediated actomyosin contractility is required for boundary formation and LC-utricle separation.*

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However, the precise roles of Lmx1a and Rock in regulating cell behaviors during boundary formation were not clearly fleshed out. For example, phenotypic analysis of Lmx1a was rather cursory; it is unclear how Lmx1a, expressed in half of the boundary domain, control boundary cell behaviors and prevent cell mixing between Lmx1a+ and Lmx1a- compartments? Well-established mechanisms and molecules for boundary formation were not investigated (e.g. differential adhesion via cadherins, cell repulsion via ephrin-Eph signaling). Moreover, within the boundary domain, it is unclear whether apical multicellular rosettes and basal constrictions are drivers of boundary formation, as boundary can still form when these cell behaviors were inhibited. Involvement of other cell behaviors, such as radial cell intercalation and oriented cell division, also warrant consideration. With these lingering questions, the mechanistic advance of the present study is somewhat incremental.

We have acknowledged the lingering questions this referee points out in our Discussion and agree that the roles of differential cell adhesion and cell intercalation would be worth exploring in further studies. Despite these remaining questions, the conceptual advances are significant, since this study provides the first evidence that a tissue boundary forms in between segregating sensory organs in the inner ear (there are only a handful of embryonic tissues in which a tissue boundary has been found in vertebrates) and highlights the evolutionary conservation of this process. This work also provides a strong descriptive basis for any future study investigating the mechanisms of tissue boundary formation in the mouse and chicken embryonic inner ear.

Reviewer #2 (Public review):

Summary:

Chen et al. describe the mechanisms that separate the common pan-sensory progenitor region into individual sensory patches, which presage the formation of the sensory epithelium in each of the inner ear organs. By focusing on the separation of the anterior and then lateral cristae, they find that long supra-cellular cables form at the interface of the pansensory domain and the forming cristae. They find that at these interfaces, the cells have a larger apical surface area, due to basal constriction, and Sox2 is down-regulated. Through analysis of Lmx1 mutants, the authors suggest that while Lmx1 is necessary for the complete segregation of the sensory organs, it is likely not necessary for the initial boundary formation, and the down-regulation of Sox2.

Strengths:

The manuscript adds to our knowledge and provides valuable mechanistic insight into sensory organ segregation. Of particular interest are the cell biological mechanisms: The authors show that contractility directed by ROCK is important for the maintenance of the boundary and segregation of sensory organs.

Weaknesses:

The manuscript would benefit from a more in-depth look at contractility - the current images of PMLC are not too convincing. Can the authors look at p or ppMLC expression

> *in an apical view? Are they expressed in the boundary along the actin cables? Does Y-27362 inhibit this expression?*
>
> *The authors suggest that one role for ROCK is the basal constriction. I was a little confused about basal constriction. Are these the initial steps in the thinning of the intervening nonsensory regions between the sensory organs? What happens to the basally constricted cells as this process continues?*

In our hands, the PMLC immunostaining gave a punctate staining in epithelial cells and was difficult to image and interpret in whole-mount preparations, which did not allow us to investigate its specific association to the actin-cable-like structures. It is a very valuable suggestion to try alternative methods of fixation to improve the quality of the staining and images in future work.

The basal constriction of the cells at the border of the sensory organs was not always clearly visible in freshly-fixed samples, and was absent in the majority of short-term organotypic cultures in control medium, which made it impossible to ascertain the role of ROCK in its formation using pharmacological approaches in vitro (see Figure 7 and corresponding Result section). On the other hand, the overexpression of a dominant-negative form of ROCK (RCII-GFP) in ovo using RCAS revealed a persistence of basal constriction in transfected cells despite a disorganisation of the boundary domain (Figure 8). We conclude from these experiments that ROCK activity is not necessary for the formation and maintenance of the basal constriction. We also remain uncertain about the exact role of this basal constriction. It could be either a cause or consequence of the expansion of the apical surface of cells in the boundary domain, it could contribute to the limitation of cell intermingling and the formation of the actin-cable-like structure at the interface of Lmx1a-expressing and non-expressing cells, and may indeed prefigure some of the further changes in cell morphology occurring in non-sensory domains separating the sensory organs (cell flattening and constrictions of the epithelial walls in between sensory organs).

> *The steps the authors explore happen after boundaries are established. This correlates with a down-regulation of Sox2, and the formation of a boundary. What is known about the expression of molecules that may underlie the apparent interfacial tension at the boundaries? Is there any evidence for differential adhesion or for Eph-Ephrin signalling? Is there a role for Notch signalling or a role for Jag1 as detailed in the group's 2017 paper?*

Great questions. It is indeed likely that some form of differential cell tension and/or adhesion participates to the formation and maintenance of this boundary, and we have mentioned in the discussion some of the usual suspects (cadherins, eph/ephrin signalling,…) although it is beyond the scope of this paper to determine their roles in this context.

As we have discussed in this paper and in our 2017 study (see also Ma and Zhang, Development, 2015 Feb 15;142(4):763-73. doi: 10.1242/dev.113662) we believe that Notch signalling is maintaining prosensory character, and its down-regulation by Lmx1a/b expression is required for the specification of the non-sensory domains in between segregating sensory organs. Although we have not tested this directly in this study, any disruption in Notch signalling would be expected to affect indirectly the formation or maintenance of the boundary domain.

> *A comment on whether cellular intercalation/rearrangements may underlie some of the observed tissue changes.*

We have not addressed this topic directly in the present study but we have included a brief comment on the potential implication of cellular intercalation and rearrangements in the discussion: “It is also possible that the repositioning of cells through medial intercalation could contribute to the straightening of the boundary as well as the widening of the nonsensory territories in between sensory patches.”

> *The change in the long axis appears to correlate with the expression of Lmx1a (Fig 5d). The authors could discuss this more. Are these changes associated with altered PCP/Vangl2 expression?*

We are not sure about the first point raised by the referee. We have quantified cell elongation and orientation in Lmx1a-GFP heterozygous and homozygous (null) mice, and our results suggest that the elongation of the cells occurs throughout the boundary domain, and is probably not dependent on Lmx1a expression (boundary cells are in fact more elongated in the Lmx1a mutant). We have not investigated the expression of components of the planar cell polarity pathway. This is a very interesting suggestion, worth exploring in further studies.

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *Lmx1a is an orthologue of apterous in flies, which is important for dorsal-ventral border formation in the wing disc. Previously, this research group has described the importance of the chicken Lmx1b in establishing the boundary between sensory and non-sensory domains in the chicken inner ear. Here, the authors described a series of cellular changes during border formation in the chicken inner ear, including alignment of cells at the apical border and concomitant constriction basally. The authors extended these observations to the mouse inner ear and showed that these morphological changes occurred at the border of Lmx1a positive and negative regions, and these changes failed to develop in Lmx1a mutants. Furthermore, the authors demonstrated that the ROCK-dependent actomyosin contractility is important for this border formation and blocking ROCK function affected epithelial basal constriction and border formation in both in vitro and in vivo systems.*
>
> *Strengths:*
>
> *The morphological changes described during border formation in the developing inner ear are interesting. Linking these changes to the function of Lmx1a and ROCK dependent actomyosin contractile function are provocative.*
>
> *Weaknesses:*
>
> *There are several outstanding issues that need to be clarified before one could pin the morphological changes observed being causal to border formation and that Lmx1a and ROCK are involved.*

We have addressed the specific comments and suggestions of the reviewer below. We wish however to point out that we do not think that ROCK activity is required for the formation or maintenance of the basal constriction at the interface of Lmx1a-expressing and nonexpressing cells (see previous answer to referee #2)

> ***Reviewer #1 (Recommendations for the authors):***
>
> *Specific comments:*
>
> *(1) Figures 1 and 2, and related text. Based on the whole-mount images shown, the anterior otocyst appeared to be a stratified epithelium with multiple cell layers. If so, it should be clarified whether the x-y view of in the "apical" and "basal" plane are from*

> *cells residing in the apical and basal layers, respectively. Moreover, it would be helpful to include a "stage 4", a later stage to show if and when basal constrictions resolve.*

In fact, at these early stages of development, the otic epithelium is “pseudostratified”: it is formed by a single layer of irregularly shaped cells, each extending from the base to the apical aspect of the epithelium, but with their nuclei residing at distinct positions along this basal-apical axis as mitotic cells progress through the cell cycle. The nuclei divide at the surface of the epithelium, then move back to the most basal planes within daughter cells during interphase. This process, known as interkinetic nuclear migration, has been well described in the embryonic neural tube and occurs throughout the developing otic epithelium (e.g. Orr, Dev Biol. 1975, 47,325-340, Ohta et al., Dev Biol. 2010 Sep 15;347(2):369–381. doi: 10.1016/j.ydbio.2010.09.002;). Consequently, the nuclei visible in apical or basal planes in x-y views belong to cells extending from the base to the apex of the epithelium, but which are at different stages of the cell cycle.

We have not included a late stage of sensory organ segregation in this study (apart from a P0 stage in the mouse inner ear, see Figure 4) since data about later stages of sensory organ morphogenesis are available in other studies, including our Mann et al. eLife 2017 paper describing Lmx1a-GFP expression in the embryonic mouse inner ear.

> *(2) Related to above, the observed changes in cell organization raised the possibility that the apical multicellular rosettes and basal constrictions observed in Stage 3 (and 2) could be intermediates of radial cell intercalations, which would lead to expansion of the space between sensory organs and thinning of the boundary domains. To see if it might be happening, it would be helpful to include DAPI staining to show the overall tissue architecture at different stages and use optical reconstruction to assess the thickness of the epithelium in the presumptive boundary domain over time.*

We agree with this referee. Besides cell addition by proliferation and/or changes in cell morphology, radial cell intercalations could indeed contribute to the spatial segregation of inner ear sensory organs (a brief statement on this possibility was added to the Discussion). It is clear from images shown in Figure 4 (and from other studies) that the non-sensory domain separating the cristae from the utricle gets flatter and its cells also enlarge as development proceeds. We do not think that DAPI staining is required to demonstrate this. Perhaps the best way to show that radial cell intercalations occur would be to perform liveimaging of the otic epithelium, but this is technically challenging in the mouse or chicken inner ear. An alternative model system might be the zebrafish inner ear, in which some liveimaging data have shown a progressive down-regulation of Jag1 expression during sensory organ segregation (and a flattening of “boundary domains”), suggesting a conservation of the basic mechanisms at play (Ma and Zhang, Development, 2015 Feb 15;142(4):763-73. doi: 10.1242/dev.113662).

> *(3) Similarly, it would be helpful to include the DAPI counterstain in Figures 4, 7, and 8 to show the overall tissue architecture.*

We do not have DAPI staining for these particular images but in most cases, Sox2 immunostaining gives a decent indication of tissue morphology.

> *(4) Figure 2(z) and Figure 4d. The arrows pointing at the basal constrictions are obstructing the view of the basement membrane area, making it difficult to appreciate the morphological changes. They should be moved to the side. Can the authors comment whether they saw evidence for radial intercalations (e.g. thinning of the boundary domain) or partial unzippering of adjoining compartments along the basal constrictions?*

The arrows in Figure 2(z) and Figure 4d have been moved to the side of the panels.

See previous comment. Besides the presence of multicellular rosettes, we have not seen direct evidence of radial cell intercalation – this would be best investigated using liveimaging. As development proceeds, the epithelial domain separating adjoining sensory organs becomes wider. The cells that compose it gradually enlarge and flatten, as can be seen for example at P0 in the mouse inner ear (Figure 4g).

> *(5) Figures 3 and 5, and related text. It should be clarified whether the measurements were all taken from the surface cells. For Fig. 3e and 5d, the mean alignment angles of the cell long axis in the boundary regions should be provided in the text.*

The sensory epithelium in the otocyst is pseudostratified, hence, the measurement was taken from the surface of all epithelial cells labelled with F-actin.

We have added histograms representing the angular distribution of the cell long axis orientations in the boundary region to Figure 3 and Figure 5 Supplementary 1. We believe that this type of representation is more informative than the numerical value of the mean alignment angles of the cell long axis for defined sub-domains.

> *(6) It would be helpful to also quantify basal constrictions using the cell skeleton analysis. In addition, it would be helpful to show x-y views of cell morphology at the level of basal constrictions in the mouse tissue, similar to the chick otocyst shown in Figure 2.*

The data that we have collected do not allow a precise quantification of basal constrictions with cell skeleton analysis, due to the generally fuzzy nature of F-actin staining in the basal planes of the epithelium. However, we have followed the referee's advice and analysed Factin staining in x-y views in the Lmx1a-GFP knock-in (heterozygous) mice. We found that the first signs of basal F-actin enrichment and multicellular actin-cable like structures at the interface of Lmx1a-positive and negative cells are visible at E11.5, and F-actin staining in the basal planes increases in intensity and extent at E13.5. (shown in new Figure 4 – Supplementary Figure 1).

> *(7) Figure 5 and related text. It would be informative to analyze Lmx1a mutants at early stages (E11-E13) to pinpoint cell behavior defects during boundary formation.*

We chose the E15 stage because it is one at which we can unequivocally recognize and easily image and analyse the boundary domain from a cytoarchitectural point of view. We recognize that it would have been worth including earlier stages in this analysis but have not been able to perform these additional studies due to time constraints and unavailability of biological material.

> *(8) Figure 5-Figure S1, the quantifications suggest that Lmx1a loss had both cellautonomous and non-autonomous effects on boundary cell behaviors. This is an interesting finding, and its implication should be discussed.*

It is well-known that the absence of Lmx1a function induces a very complex (and variable) phenotype in terms of inner ear morphology and patterning defects. It is also clear from this study that the absence of Lmx1 causes non-cell autonomous defects in the boundary domain and we have already mentioned this in the discussion: "Finally, the patterning abnormalities in Lmx1a$^{GFP/GFP}$ samples occurred in both GFP-positive and negative territories, which points at some type of interaction between Lmx1a-expressing and nonexpressing cells, and the possibility that the boundary domain is also a signalling centre influencing the differentiation of adjacent territories."

> *(9) Figure 6 and related text. To correlate myosin II activity with boundary cell behaviors, it would be important to immunolocalize pMLC in the boundary domain in whole-mount otocyst preparations from stage 1 to stage 3.*

We tried to perform the suggested immunostaining experiments, but in our hands at least, the antibody used did not produce good quality staining in whole-mount preparations. We have therefore included images of sectioned otic tissue, which show some enrichment in pMLC immunostaining at the interface of segregating organs (Figure 6).

> *(10) Figures 7 and 8. A caveat of long-term Rock inhibition is that it can affect cell proliferation and differentiation of both sensory and non-sensory cells, which would cause secondary effects on boundary formation. This caveat was not adequately addressed. For example, does Rock signaling control either the rate or the orientation of cell division to promote boundary formation? Together with the mild effect of acute Rock inhibition, the precise role of Rock signaling in boundary formation remains unclear.*

We absolutely agree that the exact function of ROCK could not be ascertained in the in vitro experiments, for the reasons we have highlighted in the manuscript (no clear effect in short term treatments, great level of tissue disorganisation in long-term treatments). This prompted us to turn to an in ovo approach. The picture remains uncertain in relation to the role of ROCK in regulating cell division/intercalation but we have been at least able to show a requirement for the maintenance of an organized and regular boundary.

> *(11) Figure 8. RCII-GFP likely also have non-autonomous effects on cell apical surface area. In 8d, it would be informative to include cell area quantifications of the GFP control for comparison.*

It is possible that some non-autonomous effects are produced by RCII-GFP expression, but these were not the focus of the present study and are not particularly relevant in the context of large patches of overexpression, as obtained with RCAS vectors.

We have added cell surface area quantifications of the control RCAS-GFP construct for comparison (Figure 8e).

> *(12) The significance of the presence of cell divisions shown in Figure 9 is unclear. It would be informative to include some additional analysis, such as a) quantify orientation of cell divisions in and around the boundary domain and b) determine whether patterns of cell division in the sensory and nonsensory regions are disrupted in Lmx1a mutants.*

These are indeed fascinating questions, but which would require considerable work to answer and are beyond the scope of this paper.

> *Minor comments:*
>
> *(1) Figure 1. It should be clarified whether e', h' and k' are showing cortical F-actin of surface cells. Do the arrowheads in i' and l' correspond to the position of either of the arrowheads in h' and k', respectively?*

The epithelium in the otocyst is pseudostratified. Therefore, images e’, h’, k’ display F-actin labelling on the surface of tissue composed of a single cell layer. We have added arrows to images e”, h”, and k” to indicate the corresponding position of z-projections and included appropriate explanation in the legend of Figure 1: “Black arrows on the side of images e”, h”, and k” indicate the corresponding position of z-projections.”

> *(2) Figure 3-Figure S1. Please mark the orientation of the images shown.*

We labelled the sensory organs in the figure to allow for recognizing the orientation.

> *(3) Figure 4. Orthogonal reconstructions should be labeled (z) to be consistent with other figures.*

We have corrected the labelling in the orthogonal reconstruction to (z).

> *(4) Figure 4g. It is not clear what is in the dark area between the two bands of Lmx1a+ cells next to the utricle and the LC. Are those cells Lmx1a negative? It is unclear whether a second boundary domain formed or the original boundary domain split into two between E15 and P0? Showing the E15 control tissue from Figure 5 would be more informative than P0.*

In this particular sample there seems to be a folding of the tissue (visible in z-reconstructions) that could affect the appearance of the projection shown in 4g. We believe the P0 is a valuable addition to the E15 data, showing a slightly later stage in the development of the vestibular organs.

> *(5) Figure 5a, e. Magnified regions shown in b and f should be boxed correspondingly.*

This figure has been revised. We realized that the previous low-magnification shown in (e) (now h) was from a different sample than the one shown in the high-magnification view. The new figure now includes the right low-magnification sample (in h) and the regions shown in the high-magnification views have been boxed.

> *(6) Figure 8f, h, j. Magnified regions shown in g, i and k should be boxed correspondingly.*

The magnified regions were boxed in Figure 8 f, h, and j. Additionally, black arrows have been placed next to images 8g'', 8i'', and 8k'' to highlight the positions of the z-projections. An appropriate explanation has also been added to the figure legend.

> *(9) Figure 8. It would be helpful to show merged images of GFP and F-actin, to better appreciate cell morphology of GFP+ and GFP- cells.*

As requested, we have added images showing overlap of GFP and F-actin channels in Figure 8.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *The PMLC staining could be improved. Two decent antibodies are the p-MLC and pp-MLC antibodies from CST. pp-MLC works very well after TCA fixation as detailed in https://www.researchsquare.com/article/rs-2508957/latest . As phalloidin does not work well after TCA fixation, affadin works very well for segmenting cells.*
>
> *If the authors do not wish to repeat the pMLC staining, the details of the antibody used should be mentioned.*

We used mouse IgG1 Phospho-Myosin Light Chain 2 (Ser19) from Cell Signaling Technology (catalogue number #3675) in our immunohistochemistry for PMLC. This is one of the two antibodies recommended by the reviewer #2. Information about this antibody has now been included in material and methods. This antibody has been referenced by many manuscripts, but unfortunately, in our hands at least, it did not perform well in whole-mount preparations.

> *A statement on the availability of the data should be included.*

We have included a statement on the data availability: "All data generated or analysed during this study is available upon request."

> ***Reviewer #3 (Recommendations for the authors):***
>
> *Outstanding issues:*
>
> *(1) Morphological description: The apical alignment of epithelial cells at the border is clear but not the upward pull of the basal lamina. Very often, it seems to be the Sox2 staining that shows the upward pull better than the F-actin staining. Perhaps, adding an anti-laminin staining to indicate the basement membrane may help.*

Indeed, the upward pull of the basement membrane is not always very clear. We performed some anti-laminin immunostaining on mouse cryosections and provide below (Figure 1) an example of such experiment. The results appear to confirm an upward displacement of the basement membrane in the region separating the lateral crista from the utricle in the E13 mouse inner ear, but given the preliminary nature of these experiments, we believe that these results do not warrant inclusion in the manuscript. The term "pull" is somehow implying that the epithelial cells are responsible for the upward movement of the basement membrane, but since we do not have direct evidence that this is the case, we have replaced "pull" by "displacement" throughout the text.

> *(2) It is not clear how well the cellular changes are correlated with the timing of border formation as some of the ages shown in the study seem to be well after the sensory patches were separated and the border was established.*

For some experiments (for example E15 in the comparison of mouse Lmx1a-GFP heterozygous and homozygous inner ear tissue; E6 for the RCAS experiments), the early stages of boundary formation are not covered because we decided to focus our analysis on the late consequences of manipulating Lmx1a/ROCK activity in terms of sensory organ segregation. The dataset is more comprehensive for the control developmental series in the chicken and mouse inner ear.

> *(3) The Lmx1a data, as they currently stand could be explained by Lmx1a being required for non-sensory development and not necessarily border formation. Additionally, the relationship between ROCK and Lmx1a was not investigated. Since the investigators have established the molecular mechanisms of Lmx1 function using the chicken system previously, the authors could try to correlate the morphological events described here with the molecular evidence for Lmx1 functioning during border formation in the same chicken system. Right now, only the expression of Sox2 is used to correlate with the cellular events, and not Lmx1, Jag1 or notch.*

These are valid points. Exploring in detail the epistatic relationships between Notch signalling/Lmx1a/ROCK/boundary formation in the chicken model would be indeed very interesting but would require extensive work using both gain and loss-of-function approaches, combined with the analysis of multiple markers (Jag1/Sox2/Lmx1b/PMLC/Factin..). At this point, and in agreement with the referee's comment, we believe that Lmx1a is above all required for the adoption of the non-sensory fate. The loss of Lmx1a function in the mouse inner ear produce defects in the patterning and cellular features of the boundary domain, but these may be late consequences of the abnormal differentiation of the nonsensory domains that separate sensory organs. Furthermore, ROCK activity does not appear to be required for Sox2 expression (i.e. adoption or maintenance of the sensory fate) since the overexpression of RCII-GFP does not prevent Sox2 expression in the chicken inner ear. This fits with a model in which Notch/Lmx1a regulate cell differentiation whilst ROCK acts independently or downstream of these factors during boundary formation.

> *Specific comments:*
>
> *(1) Figure 1. The downregulation of Sox2 is consistent between panels h and k, but not between panels e and h. The orthogonal sections showing basal constriction in h' and k' are not clear.*

The downregulation is noticeable along the lower edge of the crista shown in h; the region selected for the high-magnification view sits at an intermediate level of segregation (and Sox2 downregulation).

The basal constriction is not very clear in h, but becomes easier to visualize in k. We have displaced the arrow pointing at the constriction, which hopefully helps.

> *(2) Figure 2. Where was the Z axis taken from? One seems to be able to imagine the basal constriction better in the anti-Sox2 panel than the F-actin panel. A stain outlining the basement membrane better could help.*

Arrows have been added on the side of the horizontal views to mark the location of the zreconstruction. See our previous replies to comments addressing the upward displacement of the basement membrane.

> *(3) Figure 4*
>
> *I question the ROI being chosen in this figure, which seems to be in the middle of a triad between LC, prosensory/utricle and the AC, rather than between AC and LC. If so, please revise the title of the figure. This could also account for the better evidence of the apical alignment in the upper part of the f panel.*

We have corrected the text.

> *In this figure, the basal constriction is a little clearer in the orthogonal cuts, but it is not clear where these sections were taken from.*

We have added black arrows next to images 4c’, 4f’, and 4i’ to indicate the positions of the zprojections.

> *By E13.5, the LC is a separate entity from the utricle, it makes one wonder how well the basal constriction is correlated with border formation. The apical alignment is also present by P0, which raises the question that the apical alignment and basal restriction may be more correlated with differentiation of non-sensory tissue rather than associated with border formation.*

We agree E13.5 is a relatively late stage, and the basal constriction was not always very pronounced. The new data included in the revised version include images of basal planes of the boundary domain at E11.5, which reveal F-actin enrichment and the formation of an actin-cable-like structure (Figure 4 suppl. Fig1). Furthermore, the chicken dataset shows that the changes in cell size, alignment, and the formation of actin-cable-like structure precede sensory patch segregation and are visible when Sox2 expression starts to be downregulated in prospective non-sensory tissue (Figure 1, Figure 2). Considering the results from both species, we conclude that these localised cellular changes occur relatively early in the sequence of events leading to sensory patch segregation, as opposed to being a late consequence of the differentiation of the non-sensory territories.

> *I don't follow the (x) cuts for panels h and I, as to where they were taken from and why there seems to be an epithelial curvature and what it was supposed to represent.*

We have added black arrows next to the panels 4c', 4f', and 4i' to indicate the positions of the z-projections and modified the legend accordingly. The epithelial curvature is probably due to the folding of the tissue bordering the sensory organs during the manipulation/mounting of the tissue for imaging.

> *(4) Figure 5 The control images do not show the apical alignment and the basal constriction well. This could be because of the age of choice, E15, was a little late. Unfortunately, the unclarity of the control results makes it difficult for illustrating the lack of cellular changes in the mutant. The only take-home message that one could extract from this figure is a mild mixing of Sox2 and Lmx1a-Gfp cells in the mutant and not much else. Also, please indicate the level where (x) was taken from.*

Black arrows have been placed next to images 5e and 5l to highlight the positions of the zprojections. The stage E15 chosen for analysis was appropriate to compare the boundary domains once segregation is normally completed. We believe the results show some differences in the cellular features of the boundary domain in the Lmx1a-null mouse, and we have in fact quantified this using Epitool in Figure 5 – Suppl. Fig 1. Cells are more elongated and better aligned in the Lmx1a-null than in the heterozygous samples.

> *(5) Figure 7. I think the cellular disruption caused by the ROCK inhibitor, shown in q', is too severe to be able to pin to a specific effect of ROCK on border formation. In that regard, the ectopic expression of the dominant negative form of ROCK using RCAS approach is better, even though because it is a replication competent form of RCAS, it is still difficult to correlate infected cells to functional disruption.*

We used a replication-competent construct to induce a large patch of infection, increasing our chances of observing a defect in sensory organ segregation and boundary formation. We agree that this approach does not allow us to control the timing of overexpression, but the mosaicism in gene expression, allowing us to compare in the same tissue large regions with/without perturbed ROCK activity, proved more informative than the pharmacological/in vitro experiments.

> *(6) Figure 8. Outline the ROI of i in h, and k in j. Outline in k the comparable region in k'. In k", F-actin staining is not uniform. Indicate where (x) was taken from in K.*

The magnified regions were boxed in Figure 8 f, h, and j. Region outlined in figures k'-k" has also been outlined in corresponding region in figure k. Additionally, black arrows have been placed next to images 8g", 8i", and 8k" to highlight the positions of the z-projections. An appropriate explanation has also been added to the figure legend.

> *Minor comments:*
>
> *(1) P.18, 1st paragraph, extra bracket at the end of the paragraph.*

Bracket removed

> *(2) P.22, line 11, in ovo may be better than in vivo in this case.*

We agree, this has been corrected.

> *(3) P.25, be consistent whether it is GFP or EGFP.*

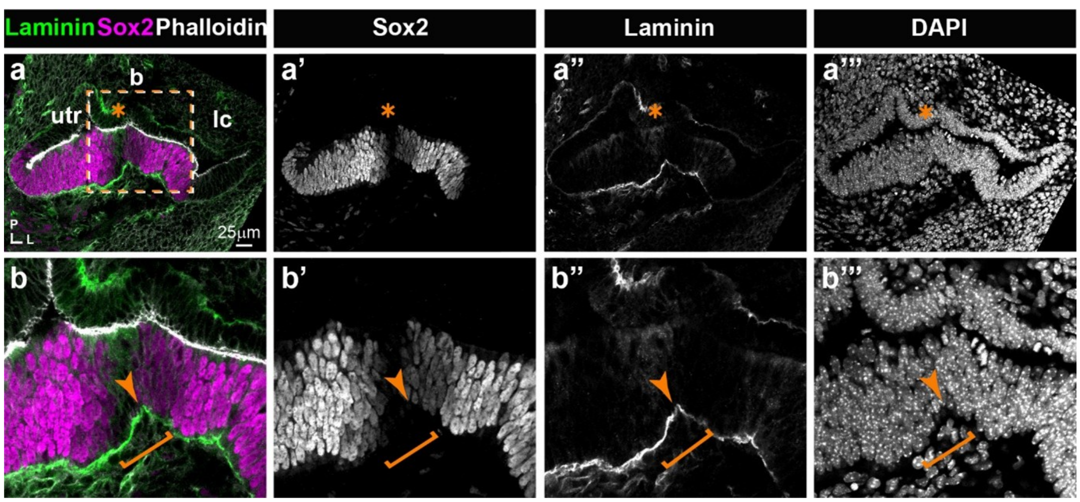
Corrected to GFP.

> *(4) P.26, line 5. Typo on "an"*

Corrected to “and”

Author response image 1.

Expression of Laminin and Sox2 in the E13 mouse inner ear. a-a’”) Low magnification view of the utricle, the lateral crista, and the non-sensory (Sox2-negative) domain separating these. Laminin staining is detected at relatively high levels in the basement membrane underneath the sensory patches. At higher magnification (b-b’”), an upward displacement of the basement membrane (arrow) is visible in the region of reduced Sox2 expression, corresponding to the “boundary domain” (bracket).



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